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DATA | ANALYTICS | CAREER

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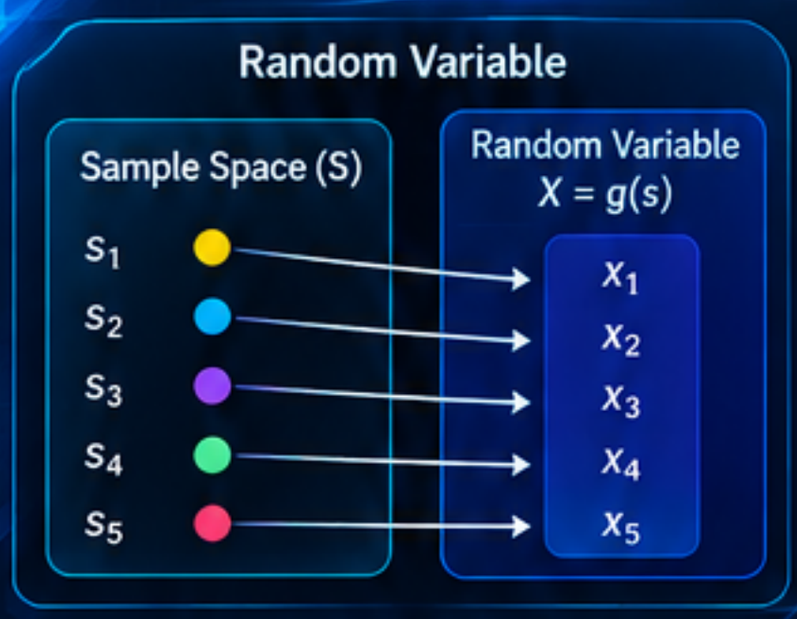
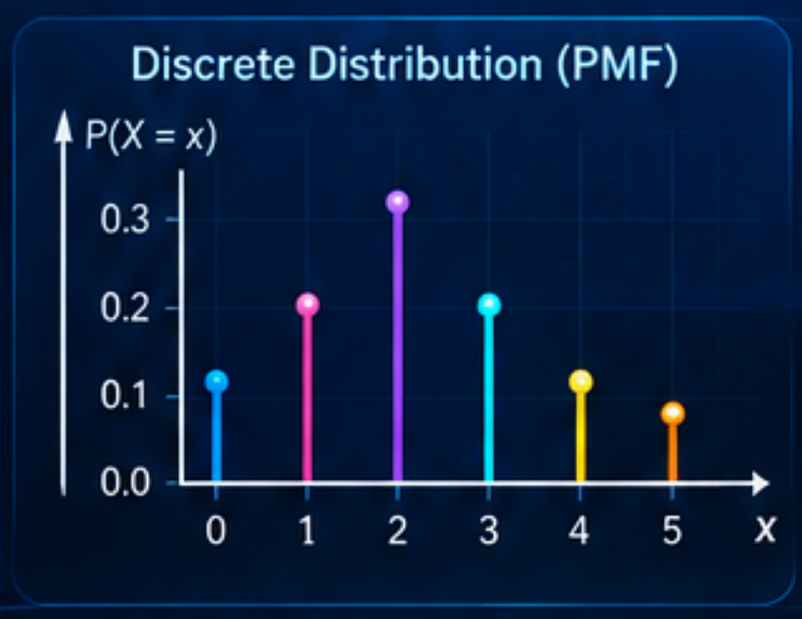
MODULE 5

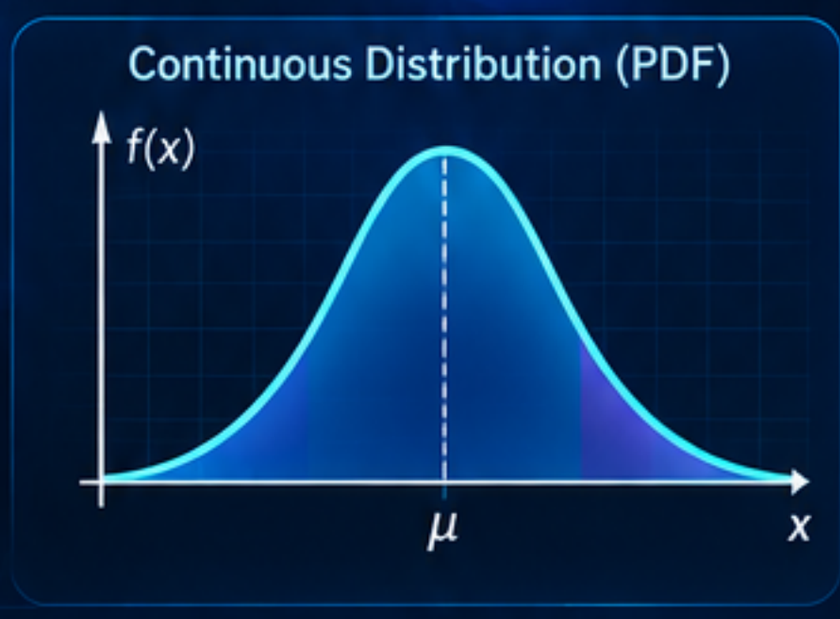
RANDOM VARIABLES & PROBABILITY DISTRIBUTIONS

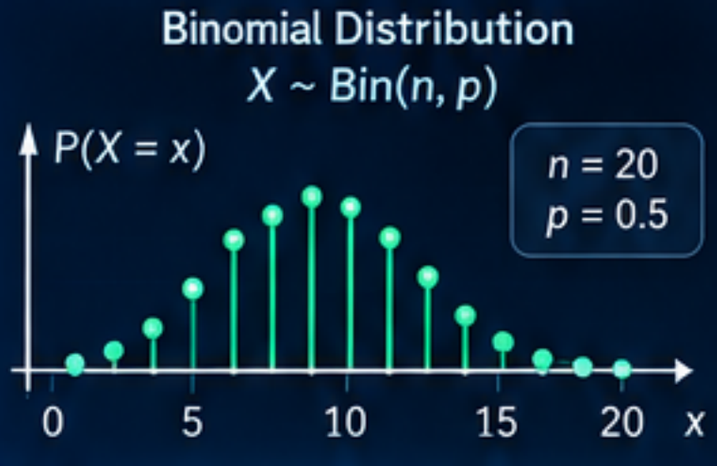
Statistics for **Data Analytics & Data Science**

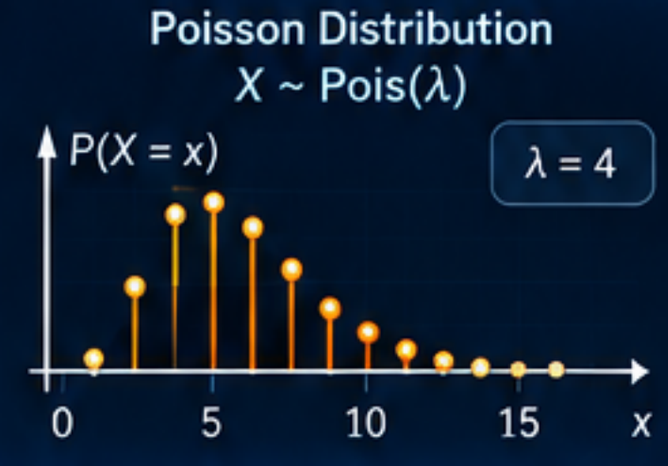
Concepts | Formulas | Examples | Excel Practice | Real-World Use Cases | Interview Concepts

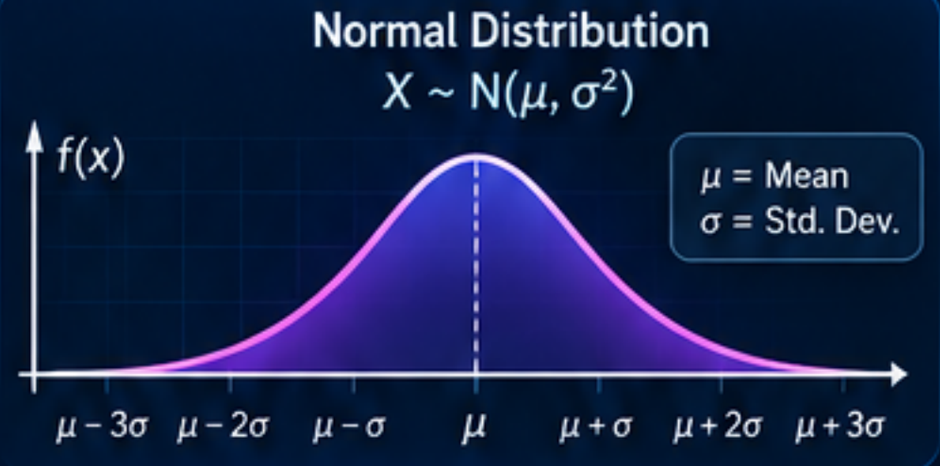












14 TOPICS IN THIS MODULE WE WILL COVER

1. Random Variable: Definition
2. Discrete Random Variables
3. Continuous Random Variables
4. Probability Mass Function (PMF)
5. Probability Density Function (PDF)
6. Expected Value (Mean of Random Variable)
7. Variance & Standard Deviation of Random Variable
8. Binomial Distribution
9. Poisson Distribution
10. Normal Distribution in Depth
11. Standard Normal Distribution
12. Properties of Normal Distribution
13. Sampling Distribution (Introduction)
14. Real-World Applications & Interview Concepts

Build Strong Concepts | Solve Real-World Problems | Hands-on Excel Practice | Prepare for Interviews | Apply in Business & Analytics

MODULE 5

1. Random Variables: Definition

A Numerical WAY to Represent Outcomes of a **Random Experiment**

What is a Random Variable?

A random variable is a numerical function that assigns a real number to each outcome of a random experiment.

> *"It converts the outcomes of a random experiment into numbers."*

Key Points

- Represents outcomes with numbers
- Can be discrete or continuous
- Values are determined by chance
- Used to quantify uncertainty
- Essential for probability distributions
- Widely used in business, finance, healthcare, manufacturing and data analytics

Why is it Important?

- Helps in calculating probabilities
- Used to find expected value, variance, etc.
- Forms the basis for probability distributions
- Widely used in real-world decision making
- Essential in statistics, machine learning and risk analysis

Simple Example

When we roll a fair six-sided die, we can define a random variable X as the number that appears on the die.

$$X(1) = 1$$
$$X(2) = 2$$
$$X(3) = 3$$
$$X(4) = 4$$
$$X(5) = 5$$
$$X(6) = 6$$

Formal Definition

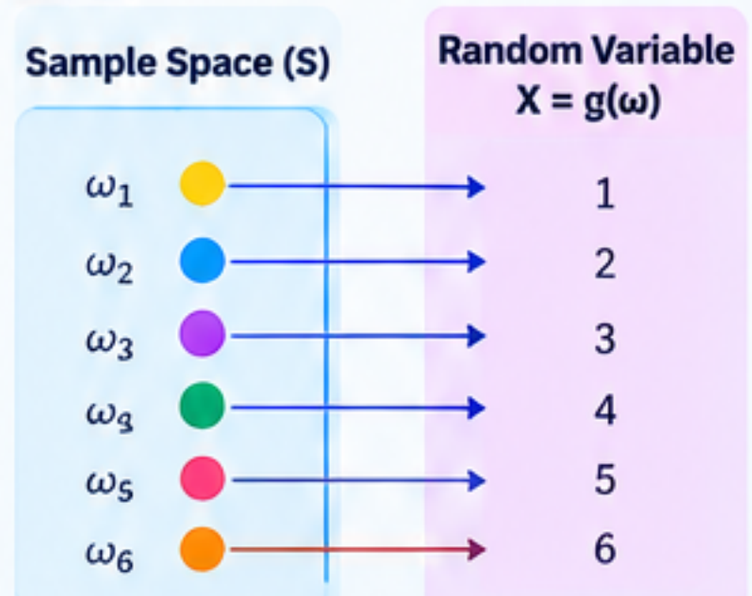
A random variable (X) is a function that maps each outcome (ω) in the sample space (S) to a real number.

$$X : S \to \mathbb{R}$$
or
$$X = g(\omega)$$

Where:

- S = sample space (set of all outcomes)
- ω = an outcome
- $X(\omega)$ = numerical value assigned to that outcome

Visual Representation



Types of Random Variables

Discrete Random Variable

- Takes countable (finite or infinite) number of values.
- Usually results from counting.
- Examples: Number of heads in 3 coin tosses, number of customers in a day.

Continuous Random Variable

- Takes any value in a given interval of real numbers.
- Usually results from measurement.
- Examples: Height of a person, time taken for delivery, weight of a product.

Real-World Use Cases

E-commerce Number of orders per day	**Finance** Daily returns of a stock	**Healthcare** Waiting time in a hospital
Manufacturing Number of defective items in a batch	**Marketing** Number of clicks on an ad	**Operations** Delivery time (in hours)

Excel Implementation

Discrete Example: Coin Tosses

Trial	Outcome	X (Value)
1	H	1
2	T	0
3	H	1
4	H	1
5	T	0

Here $X = 1$ if Head, 0 if Tail.

Continuous Example: Customer Wait Time

Customer	Wait Time (Minutes)
1	2.5
2	3.1
3	0.8
4	5.2
5	1.6

Here X represents waiting time (can take any real value).

Common Interview Concepts

1. What is a random variable?
2. Difference between discrete and continuous random variables?
3. Give examples of random variables in real life.
4. Can a random variable take negative values?
5. Is a random variable always a probability?
6. What is the range of a random variable?
7. How is a random variable related to a probability distribution?

MODULE 5

2. Discrete Random Variables

Random Variables with Countable Possible Values

What is a Discrete Random Variable?

A discrete random variable is a random variable that can take a countable number of distinct values. These values are usually obtained from counting.

> *A discrete random variable represents the outcome of a random experiment where the possible values are countable (finite or infinite)."*

Key Points

- Can take countable distinct values.
- Values are usually obtained from counting.
- Can be finite or infinite.
- Described using a probability mass function (PMF).
- Used to model many real-world situations.
- Common distributions: Binomial, Poisson, Geometric, etc.

Characteristics

- Takes only specific separated values (like 0, 1, 2, 3, ...).
- No decimals or continuous values between points.
- Often represents counts.
- Can be finite (e.g., die outcomes) or infinite (e.g., number of calls).
- Used with Probability Mass Function (PMF).
- Easier to visualize using tables and bar charts.

Examples

- Number of heads in 3 coin tosses (0, 1, 2, 3)
- Number of defective items in a batch
- Number of customers visiting a store per hour
- Number of goals in a football match
- Number of emails received in a day
- Number of accidents at a junction
- Number of calls to a customer care center

Mathematical Representation

A discrete random variable X maps each outcome ω in the sample space S to a real number.

$$X : S \to \mathbb{R}$$

or

$$X = g(\omega)$$

Where:

- S = sample space (set of all outcomes)
- ω = an outcome
- $X(\omega)$ = numerical value assigned to that outcome

Visual Representation

Discrete Random Variable (Bar Chart)

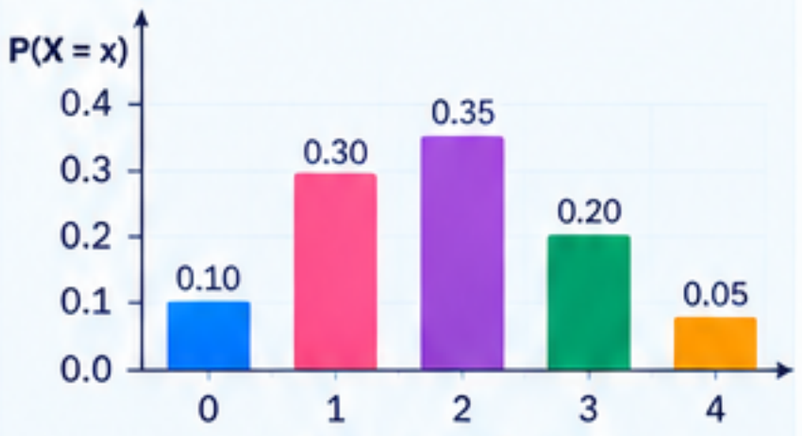


Illustration – Die Roll

Let X be the number obtained when a fair die is rolled.

Outcome (ω)	X(ω)
1	1
2	2
3	3
4	4
5	5
6	6

X can take values {1, 2, 3, 4, 5, 6} which are countable, so X is a **discrete** random variable.

Illustration – Customer Arrivals

Let X be the number of customers arriving at a store in 1 hour.

Possible Value of X (Number of Customers)	Probability P(X = x)
0	0.10
1	0.25
2	0.30
3	0.20
4	0.10
5	0.05

X takes countable values {0, 1, 2, 3, 4, 5}, so it is a **discrete random variable.**

Real-World Use Cases

Retail	Number of customers visiting a store per day
Manufacturing	Number of defective items in a batch
Telecom	Number of calls received per hour
Marketing	Number of email responses in a campaign
Sports	Number of goals in a match
Traffic	Number of accidents at a junction

Excel Implementation

Example 1: Number of Heads in 3 Tosses (Simulation)

Trial	Toss 1	Toss 2	Toss 3	X (No. of Heads)
1	H	T	H	2
2	T	T	H	1
3	H	H	H	3
4	T	T	T	0
5	H	T	T	1

Example 2: Frequency Table & Probability

X (Heads)	Frequency	Probability
0	15	0.15
1	30	0.30
2	35	0.35
3	20	0.20
Total	**100**	**1.00**

$$\text{Probability} = \frac{\text{Frequency}}{\text{Total}}$$

Common Interview Concepts

1. What is a discrete random variable?
2. Give 3 real-life examples.
3. Can a discrete random variable take decimal values? Why or why not?
4. How is a discrete random variable different from a continuous random variable?
5. Which probability distribution is commonly used for count data?
6. How do we represent a discrete random variable mathematically?

MODULE 5

3. Continuous Random Variables

Random Variables that can take Any Value in a Given Interval

What is a Continuous Random Variable?

A continuous random variable is a random variable that can take any real number value within a given interval or range. It can take infinitely many possible values.

> *A continuous random variable can take any value, including decimals, within a given range.*

Key Points

- Can take any real number value within an interval.
- Has infinitely many possible values.
- Usually obtained from measurement.
- Probability at a particular point is 0 (e.g., $P(X = 5) = 0$).
- Described using a Probability Density Function (PDF).
- Common examples: height, weight, time, temperature, income, etc.

Characteristics

- Values are not countable.
- Can take fractional or decimal values.
- Usually measured (not counted).
- Probability at a single point is zero.
- We calculate probability over an interval (range).
- Represented using a continuous curve.
- Key concept: Probability Density Function (PDF).

Real-Life Examples

- Height of a person
- Weight of a product
- Time taken to complete a task
- Temperature in a city
- Distance travelled by a vehicle
- Amount of money spent by a customer
- Waiting time at a bank or hospital
- Life time of a machine

Mathematical Representation

A continuous random variable X can take any value in an interval (a, b) or even from $-\infty$ to $+\infty$.

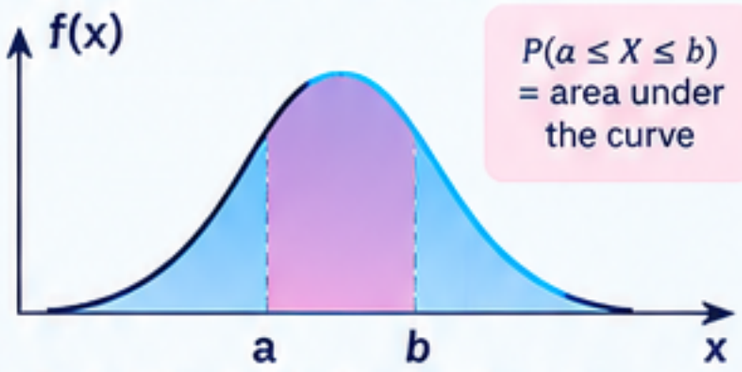
$$X : S \to \mathbb{R}$$
or
$$X = g(\omega)$$

Where:

- S = sample space (set of all outcomes)
- ω = an outcome
- $X(\omega)$ = numerical value assigned to that outcome (can be any real number)

Visual Representation

A continuous random variable is represented by a smooth curve.



Probability is calculated as the area under the curve, not at a single point.

Example – Height of Students

Let X be the height (in cm) of a student in a college. X can take any value, such as 165.2, 170.5, 171.8, etc. within a range.

Student	Height (cm)
1	165.2
2	170.5
3	168.9
4	172.3
5	169.7
...	...

Height can be any value in a range (e.g., 150 to 200 cm), so it is a continuous random variable.

Example – Waiting Time at a Bank

Let X be the waiting time (in minutes) for a customer at a bank. X can take any real value (e.g., 2.5, 3.2, 4.8 minutes, etc.).

Customer	Waiting Time (Minutes)
1	2.4
2	3.1
3	5.6
4	1.8
5	4.3
...	...

Waiting time can be any value in a range (e.g., 0 to 30 minutes), so it is a continuous random variable.

Comparison: Discrete vs Continuous

Discrete Random Variable	Continuous Random Variable
Takes countable values (like 0, 1, 2, 3, ...)	Takes any real value in an interval (like 0.5, 1.2, 3.8, ...)
Finite or infinite countable values	Infinitely many values
Usually obtained from counting	Usually obtained from measurement
Probability at a point can be > 0	Probability at a single point is 0 ($P(X = x) = 0$)
Uses PMF	Uses PDF
Examples: number of customers, number of calls	Examples: height, weight, time, temperature

Excel Implementation (Simulation)

Generate Random Continuous Values (Uniform Distribution)

Random Number (0 to 1)	Height (150 to 200 cm)
0.12	156.0
0.56	178.0
0.23	161.5
0.89	194.5
0.37	168.5

Formula (in B2): `=150 + (200-150)*A2` Generates a random height between 150 and 200 cm.

Calculate Probability in an Interval (Using Normal)

Suppose heights are normally distributed with mean = 170 cm and standard deviation = 10 cm.

Question	Excel Formula
$P(X \le 165)$	=NORM.DIST(165,170,10,TRUE)
$P(X \ge 180)$	=1 - NORM.DIST(180,170,10,TRUE)
$P(165 \le X \le 180)$	=NORM.DIST(180,170,10,TRUE) - NORM.DIST(165,170,10,TRUE)

We calculate probability over an interval, not at a single value.

Common Interview Concepts

1. What is a continuous random variable?
2. Give 3 real-life examples.
3. Why is the probability at a single point zero?
4. How is a continuous random variable different from a discrete random variable?
5. How do we calculate probability for a continuous random variable?
6. What function is used to describe a continuous random variable?
7. Which probability distributions are continuous?
8. Give business use cases of continuous random variables.

MODULE 5

4. Probability Mass Function (PMF)

Function that Assigns Probabilities to the Values of a Discrete Random Variable

What is a PMF?

A Probability Mass Function (PMF) gives the probability for each possible value of a discrete random variable.

> *It tells us the probability of each individual outcome of a discrete random variable.*

Key Points

- Used for discrete random variables.
- Gives probability for each possible value.
- Notation: $P(X = x)$
- All probabilities are between 0 and 1 $(0 \le P(X = x) \le 1)$
- Sum of all probabilities is 1 $\sum P(X = x) = 1$
- Helps in calculating mean, variance and other properties.

Properties of PMF

1. $P(X = x) \ge 0$ for all x
2. $P(X = x) \le 1$ for all x
3. $\sum P(X = x) = 1$
4. Defined only for discrete random variables
5. Completely describes the probability distribution
6. Used to calculate mean, variance and other statistical measures

Mathematical Definition

A function $p(x)$ is a probability mass function of a discrete random variable X if it satisfies:

$$p(x) = P(X = x)$$

1. $p(x) \ge 0$ for all x
2. $\sum p(x) = 1$

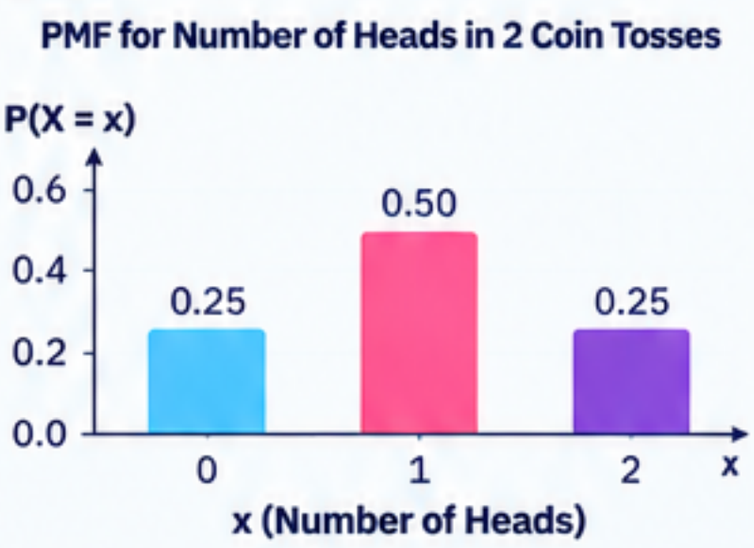
Example – Coin Tosses

Let X be the number of heads in 2 coin tosses. X can take values 0, 1, 2.

x (Number of Heads)	P(X = x)
0	1/4 = 0.25
1	2/4 = 0.50
2	1/4 = 0.25

$\sum P(X = x) = 0.25 + 0.50 + 0.25 = 1$

Visual Representation



Another Example – Dice

Let X be the outcome when a fair die is rolled. X can take values 1, 2, 3, 4, 5, 6.

x (Outcome)	P(X = x)
1	1/6 = 0.167
2	1/6 = 0.167
3	1/6 = 0.167
4	1/6 = 0.167
5	1/6 = 0.167
6	1/6 = 0.167

This is a **discrete uniform distribution** where each outcome is equally likely.

Real-Life Use Cases

- Number of defects in a batch (0, 1, 2, ...)
- Number of purchases per customer in a day
- Number of calls to a call center in an hour
- Number of emails received in a day
- Number of machines that fail in a week
- Number of customers visiting a store per day

PMF vs PDF

Aspect	PMF (Discrete)	PDF (Continuous)
Used for	Discrete random variables	Continuous random variables
Values	Countable values (0, 1, 2, ...)	Any real value in an interval
Notation	$P(X = x)$	$f(x)$
Probability	$P(X = x) \ge 0$	$f(x) \ge 0$
Total Area/Sum	$\sum P(X = x) = 1$	$\int_{-\infty}^{\infty} f(x)\,dx = 1$
At a point	P(X = x) can be ≥ 0	$P(X = x) = 0$
Graph	Bar chart (discrete points)	Smooth curve

Excel Implementation

Example: PMF for Dice using Formulas

Outcome (x)	Frequency	Probability P(X = x)
1	1	=B2/6
2	1	=B3/6
3	1	=B4/6
4	1	=B5/6
5	1	=B6/6
6	1	=B7/6
Total	6	**=SUM(C2:C7)**
		1.000

Example: Custom Discrete Distribution

x	P(X = x)
0	0.10
1	0.20
2	0.40
3	0.20
4	0.10
Total	**1.00**

In Excel, we can use SUM() to check if total probability equals 1.

Common Interview Concepts

1. What is a Probability Mass Function (PMF)?
2. State the properties of a PMF.
3. How is a PMF different from a PDF?
4. Give a real-life example of a PMF.
5. How do you verify if a given function is a valid PMF?
6. Can the probability at a point be greater than 1?
7. Why is the sum of all probabilities equal to 1?
8. How is the PMF used to calculate the expected value and variance?

MODULE 5

5. Probability Density Function (PDF)

Function that Describes the Probability Distribution of a Continuous Random Variable

What is a Probability Density Function (PDF)?

A Probability Density Function (PDF) describes the probability distribution of a continuous random variable. It shows how the probability is distributed over a range of values.

> *The PDF does not give the probability at a single point, but it helps us find the probability over an interval.*

Key Points

- Used for continuous random variables
- $f(x)$ represents the relative likelihood of a value
- Probability is calculated as the area under the curve
- $f(x) \geq 0$ for all x
- Total area under the curve is 1
- Probability at a single point is 0
- Widely used in real-world analysis

Properties of PDF

1. $f(x) \geq 0$ for all x
2. Total area under the curve = 1
   $\int_{-\infty}^{\infty} f(x)\,dx = 1$
3. $P(a \leq X \leq b) = \int_a^b f(x)\,dx$
4. Probability at a single point is 0
   $P(X = x) = 0$
5. Describes the shape of distribution
6. Helps in finding probabilities over an interval

Mathematical Definition

A probability density function $f(x)$ is a function that satisfies:

1. $f(x) \geq 0$ for all $x \in \mathbb{R}$
2. $\int_{-\infty}^{\infty} f(x)\,dx = 1$
3. $P(a \leq X \leq b) = \int_a^b f(x)\,dx$
4. $P(X = x) = 0$

Example 1 – Uniform Distribution

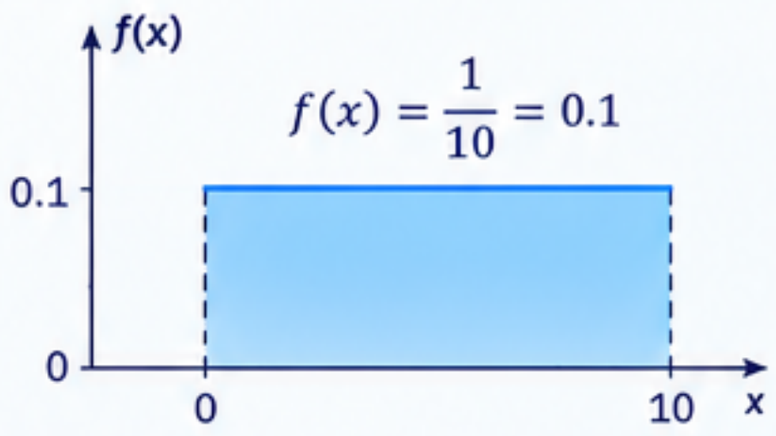
A continuous random variable X is uniformly distributed between a and b.

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ otherwise \end{cases}$$

Example: Suppose $X \sim U(0, 10)$.

$$f(x) = \begin{cases} \frac{1}{10}, & 0 \leq x \leq 10 \\ 0, & otherwise \end{cases}$$

Graph – Uniform Distribution

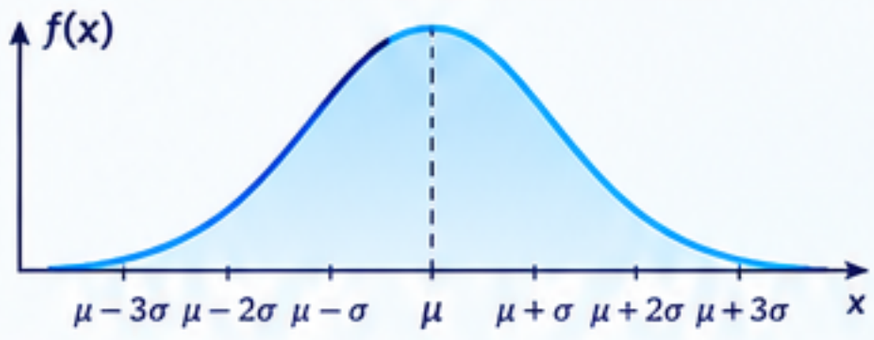


The area under the curve from 0 to 10 is 1.
This means **total probability** = 1.

Example 2 – Normal Distribution (PDF)

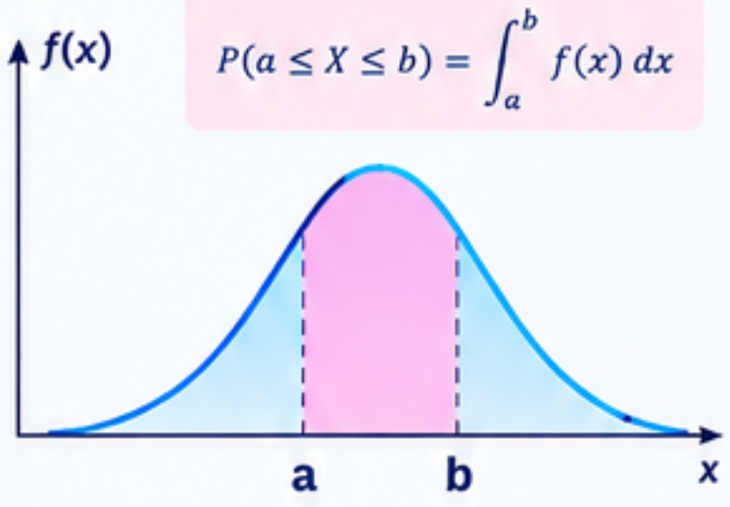
For a normal distribution with mean μ and standard deviation σ, the PDF is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



The curve is symmetric around the mean (μ) and the total area under the curve is 1.

Area Under the Curve Represents Probability



The probability of a continuous random variable lying between **a and *b* is the area** under the curve between **a and b**.

Real-World Use Cases

Height of people
Heights are continuous and follow approximately a normal distribution.

Weight of products
Product weights are continuous (random variation in manufacturing).

Time taken
Delivery time, waiting time, processing time are continuous.

Temperature
Temperature readings are continuous.

Income
Income levels are continuous.

Speed
Vehicle speed is a continuous variable.

Excel Implementation (Normal Distribution PDF)

We can calculate the PDF value in Excel using the NORM.DIST function.

x (Value)	Formula	PDF f(x) (μ = 170, σ = 10)
150	=NORM.DIST(A2,170,10,FALSE)	0.0054
160	=NORM.DIST(A3,170,10,FALSE)	0.0242
170	=NORM.DIST(A4,170,10,FALSE)	0.0399
180	=NORM.DIST(A5,170,10,FALSE)	0.0242
190	=NORM.DIST(A6,170,10,FALSE)	0.0054

Common Interview Concepts

1. What is a Probability Density Function (PDF)?
2. What is the difference between PMF and PDF?
3. Why is the probability at a **single point zero** for a continuous random variable?
4. How do we find the probability between two values (*a*, *b*)?
5. What is the total area under a PDF?
6. Give real-life examples of variables that follow a continuous distribution.
7. Which function is used in Excel to calculate the normal PDF?
8. How does the PDF help in data analysis and decision making?

MODULE 5

6. Expected Value (Mean of Random Variable)

Long-Run Average Outcome of a Random Variable

What is Expected Value?

The expected value (mean) of a random variable is the long-run average value we would expect to obtain if the experiment is repeated many times.

> *It is the weighted average of all possible values, where the weights are the probabilities of those values.*

Key Points

- Represents the long-run average.
- For discrete random variables, $E(X) = \sum xP(x)$.
- For continuous random variables, $E(X) = \int x f(x)\,dx$.
- Considers both the value and its probability.
- Very useful in business, finance, risk analysis and decision making.

Properties of Expected Value

1. $E(X)$ is a single numerical value.
2. Can be positive, negative or zero.
3. $E(aX + b) = aE(X) + b$ (linearity property).
4. For a constant c, $E(c) = c$.
5. If $X \geq 0$, then $E(X) \geq 0$.
6. Represents the centre (average) of the distribution.
7. Widely used in comparison of options and expected outcomes.

Formula – Discrete Random Variable

$$E(X) = \sum xP(x)$$

Where:

- x = possible value of the random variable
- $P(x)$ = probability of that value

Formula – Continuous Random Variable

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

Where:

- x = value of the random variable
- $f(x)$ = probability density function (PDF)

Interpretation

- It is the theoretical average value.
- It may not be one of the actual possible values.
- Over many repetitions, the sample mean approaches the expected value.
- Useful for comparing different options (e.g., games, investments, decisions).

Example 1 – Discrete (Fair Die)

Let X be the outcome when a fair die is rolled.

x	P(X = x)
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

$$E(X) = \frac{1(1/6)}{(1/6} + \frac{2(1/6)}{(1/6} + \frac{4(1/6)}{(1/6} + 5(1/6) + 6(1/6 = 3.5$$

The expected value of a fair die is **3.5**, which means in the long run, the average outcome will be 3.5.

Example 2 – Biased Coin

Let $X = 1$ if Head, 0 if Tail.
P(Head) = 0.7, P(Tail) = 0.3

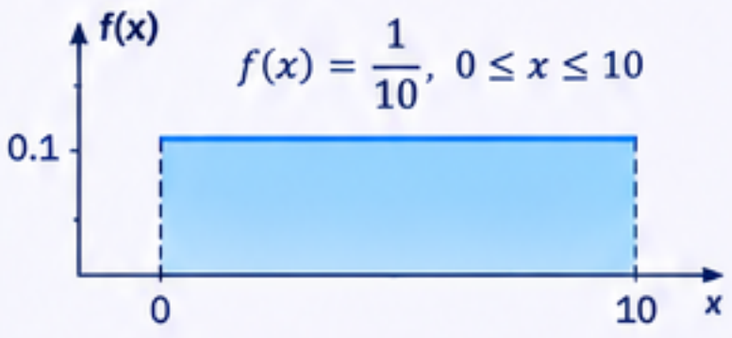
x	P(x)
1	0.7
0	0.3

$$E(X) = 1(0.7) + 0(0.3) = 0.7$$

The expected value is 0.7, which means on average, we will get 0.7 heads per toss in the long run.

Example 3 – Continuous (Uniform Distribution)

Let $X \sim U(0, 10)$ (uniform distribution between 0 and 10).



$$E(X) = \int_0^{10} x \cdot \left(\frac{1}{10}\right) dx = \frac{1}{10}\left[\frac{x^2}{2}\right]_6^{10} = \frac{10}{2} = 5$$

The expected value is 5, which is the midpoint of the interval (0, 10).

Excel Implementation

Example: Expected Value using formulas (Discrete)

Outcome (x)	Probability P(x)	x × P(x)
1	0.10	0.10
2	0.20	0.40
3	0.30	0.90
4	0.20	0.80
5	0.10	0.50
6	0.10	0.60
Total	**1.00**	**3.30**

Formula in Excel:
=SUMPRODUCT(A2:A7, B2:B7)

This gives the **expected value** $E(X) = 3.30$

Common Interview Concepts

1. What is the expected value of a random variable?
2. How is expected value different from the mean?
3. Find the expected value for a fair coin / fair die.
4. Can the expected value be negative?
5. State the linearity property of expected value.
6. Give a real-life example of expected value.
7. How is expected value used in decision making?
8. What is the expected value of a uniform distribution U(a, b)?

MODULE 5

7. Variance & Standard Deviation

Measuring the Spread or Dispersion of a Random Variable

What are Variance & Standard Deviation?

Variance and standard deviation measure the spread or dispersion of a random variable's values around its expected value (mean).

> *They tell us how much the values typically differ from the mean.*

Key Points

- Variance measures the average squared deviation from the mean.
- Standard deviation is the square root of variance.
- Both are always non-negative.
- A higher value indicates more spread in the data.
- Used for both discrete and continuous random variables.
- Important in risk analysis, forecasting and decision making.

Why Do We Use It?

1. Helps us understand the variability or risk.
2. Determines how spread out the outcomes are.
3. Used in finance (e.g., portfolio risk).
4. Used in quality control (product consistency).
5. Used in performance analysis (e.g., sales variation).
6. Used in comparing different options or strategies.

Variance Formula – Discrete Random Variable

$$Var(X) = \sigma^2 = \sum (x - \mu)^2 P(x)$$

Where:

- x = value of the random variable
- μ = expected value (mean)
- $P(x)$ = probability of that value

Variance Formula – Continuous Random Variable

$$Var(X) = \sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)\,dx$$

Where:

- x = value of the random variable
- μ = expected value (mean)
- $f(x)$ = probability density function (PDF)

Standard Deviation

Standard deviation is the square root of the variance.

$$\sigma = \sqrt{Var(X)} = \sqrt{\sigma^2}$$

Properties:

1. Always non-negative: $\sigma \geq 0$
2. Measured in the same units as data
3. If σ is small → values are close to mean
4. If σ is large → values are more spread out

Example 1 – Discrete Random Variable

Let X be the number obtained when a fair die is rolled (values 1 to 6).

x	P(x)	$(x - \mu)$	$(x - \mu)^2$	$(x - \mu)^2 P(x)$
1	1/6	-3.5	12.25	2.0417
2	1/6	-2.5	6.25	1.0417
3	1/6	-1.5	2.25	0.3750
4	1/6	0.5	0.25	0.0417
5	1/6	1.5	2.25	0.3750
6	1/6	2.5	6.25	1.0417
Sum	**1.00**	-	-	**4.9167**

$$\mu = E(X) = 3.5$$
$$Var(X) = 4.9167$$
$$\sigma = \sqrt{4.9167} = 2.22$$

Example 2 – Continuous Random Variable (Uniform Distribution)

Let $X \sim U(0, 10)$ (uniform distribution between 0 and 10).

Mean:

$$\mu = \frac{a + b}{2} = \frac{0 + 10}{2} = 5$$

Variance:

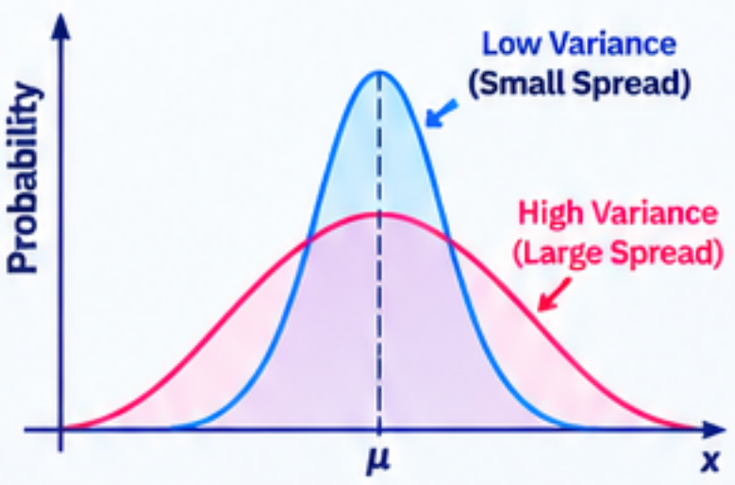
$$\sigma^2 = \frac{b - a)^2}{12} = \frac{10 - 0)^2}{12} = \frac{100}{12}$$

Standard Deviation:

$$\sigma = \sqrt{8.333} = 2.887$$

A uniform distribution has constant spread, and its standard deviation measures how far the values lie from the mean (5).

Visual Comparison



Low standard deviation → values are close to the mean (less spread).

High standard deviation → values are far from the mean (more spread).

Excel Implementation (Discrete Example)

We can calculate variance and standard deviation in Excel.

x	P(x)	$(x - \mu)^2$	$(x - \mu)^2 P(x)$
1	0.1667	12.25	2.0417
2	0.1667	6.25	1.0417
3	0.1667	2.25	0.3750
4	0.1667	0.25	0.0417
5	0.1667	2.25	0.3750
6	0.1667	6.25	1.0417
Total	**1.0000**	-	**4.9167**

Excel Formula (for variance):
`=SUMPRODUCT(C2:C7,B2:B7)`

Excel Formula (for standard deviation):
`=SQRT(SUMPRODUCT(C2:C7,B2:B7))`

Common Interview Concepts

1. What do variance and standard deviation measure?
2. Why do we square the deviations in variance?
3. How is standard deviation related to variance?
4. Can variance or standard deviation be negative?
5. Which has a larger standard deviation: a normal distribution with $\sigma = 10$ or $\sigma = 5$?
6. Give real-life examples where standard deviation is used.
7. How do variance and standard deviation help in decision making?
8. What is the variance of a constant random variable?

MODULE 5

8. Binomial Distribution

Distribution for a Fixed Number of Independent Trials with Two Outcomes

What is Binomial Distribution?

The binomial distribution gives the probability of getting a certain number of successes in a fixed number of independent trials, where each trial has two possible outcomes (success or failure) with a constant probability of success.

> *It is used when we perform the same experiment multiple times, and each trial has only two outcomes: success or failure.*

Key Points

- Fixed number of trials (n).
- Each trial has two outcomes: success or failure.
- Trials are independent.
- Probability of success (p) is constant for each trial.
- Random variable X follows a binomial distribution: $X \sim Bin(n, p)$.
- Mean $= np$
- Variance $= np(1-p)$
- Widely used in quality control, marketing, risk analysis, etc.

Conditions for Binomial Distribution

1. Fixed number of trials (n).
2. Each trial has only two outcomes (success or failure).
3. Probability of success (p) is constant for all trials.
4. Trials are independent.
5. We are interested in the number of successes (X).

Probability Formula

If $X \sim Bin(n, p)$, then

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

Where:

- n = number of trials
- x = number of successes (0, 1, 2, ..., n)
- p = probability of success
- $(1-p)$ = probability of failure

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

Mean and Variance

If $X \sim Bin(n, p)$, then

Mean: $E(X) = np$

Variance: $Var(X) = np(1-p)$

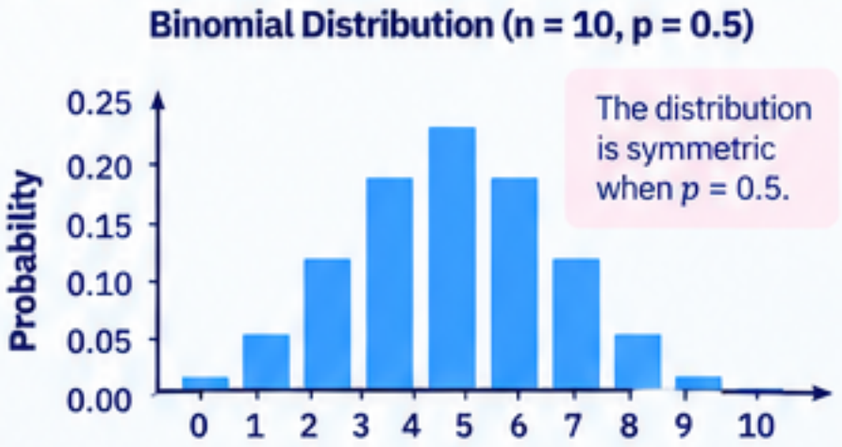
Standard Deviation:

$$\sigma = \sqrt{np(1-p)}$$

Where:

- n = number of trials
- p = probability of success

Visual Representation



For n = 10 and $p = 0.5$, the highest probability is around 5 successes.

Example 1 – Quality Control

A company manufactures light bulbs, and 5% are defective. From a sample of 10 bulbs, find the probability of getting exactly 2 defective bulbs.

Here, $n = 10, \quad p = 0.05, \quad x = 2$

$$P(X = 2) = \binom{10}{2}(0.05)^2(0.95)^8$$
$$= 45 \times 0.0025 \times 0.6634$$
$$= 0.0741$$

There is a **7.41%** chance of getting exactly 2 defective bulbs in a sample of 10.

Example 2 – Customer Purchases

A store knows that 30% of customers make a purchase. If 8 customers visit the store, find the probability that exactly 3 make a purchase.

Here, $n = 8, \quad p = 0.30, \quad x = 3$

$$P(X = 3) = \binom{8}{3}(0.3)^3(0.7)^5$$
$$= 56 \times 0.027 \times 0.16807$$
$$= 0.2537$$

There is a **25.37%** chance that exactly 3 out of 8 customers make a purchase.

Example 3 – Probability Table (n = 6, p = 0.4)

Find the probability distribution for $X \sim Bin(6, 0.4)$.

x	P(X = x)
0	0.0467
1	0.1866
2	0.3110
3	0.2765
4	0.1382
5	0.0369
6	0.0041

Mean:
$E(X) = np = 6 \times 0.4 = 2.4$

Variance:
$Var(X) = np(1-p)$
$= 6 \times 0.4 \times 0.6 = 1.44$

Standard Deviation:

$$\sigma = \sqrt{1.44} = 1.20$$

The most likely number of successes is 2 (highest probability).

Excel Implementation

We can calculate binomial probabilities in Excel using the BINOM.DIST function.

x	Formula	P(X = x)
0	=BINOM.DIST(0,6,0.4,FALSE)	0.0467
1	=BINOM.DIST(1,6,0.4,FALSE)	0.1866
2	=BINOM.DIST(2,6,0.4,FALSE)	0.3110
3	=BINOM.DIST(3,6,0.4,FALSE)	0.2765
4	=BINOM.DIST(4,6,0.4,FALSE)	0.1382
5	=BINOM.DIST(5,6,0.4,FALSE)	0.0369
6	=BINOM.DIST(6,6,0.4,FALSE)	0.0041

Cumulative Probability (At most x successes)

=BINOM.DIST(x, n, p, TRUE)

Mean and Variance in Excel

Mean = $n * p$
Variance = $n * p * (1 - p)$
Standard Deviation = SQRT($n * p * (1 - p)$)

Common Interview Concepts

1. What is a binomial distribution?
2. State the conditions for a binomial distribution.
3. Write the formula for the binomial probability.
4. How do you calculate the mean and variance?
5. Give a real-life example of a binomial distribution.
6. What happens to the shape when $p = 0.5$?
7. How is a binomial distribution different from a normal distribution?
8. When can we use normal approximation for a binomial distribution? ($np \geq 5$ and $n(1-p) \geq 5$)

MODULE 5

9. Poisson Distribution

Distribution for the Number of Events in a Fixed Interval

What is Poisson Distribution?

The Poisson distribution gives the probability of a number of events occurring in a fixed interval of time, space, or other unit, when the events happen independently and at a constant average rate.

> *It is used to model rare events that occur randomly over a fixed interval, such as the number of customer calls, defects, or arrivals.*

Key Points

- Models the number of events in a fixed interval.
- Events occur independently.
- Average rate of occurrence (λ) is constant.
- Random variable X follows a Poisson distribution: $X \sim Poisson(\lambda)$
- Mean = λ
- Variance = λ
- Only takes non-negative integer values (0, 1, 2, ...).
- Widely used for rare events and count data.

Conditions for Poisson Distribution

1. Events occur one at a time (independently).
2. The average rate (λ) is constant over the interval.
3. The probability of more than one event in a very small interval is negligible.
4. Events are independent.
5. We count the number of events in a fixed interval (time, space, etc.).

Probability Formula

If $X \sim Poisson(\lambda)$, then

$$P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

Where:

- λ = average number of events in the interval ($\lambda > 0$)
- x = number of events (0, 1, 2, ...)
- e = 2.718 (Euler's number)
- $x!$ = factorial of x

Mean and Variance

If $X \sim Poisson(\lambda)$, then

Mean: $E(X) = \lambda$

Variance: $Var(X) = \lambda$

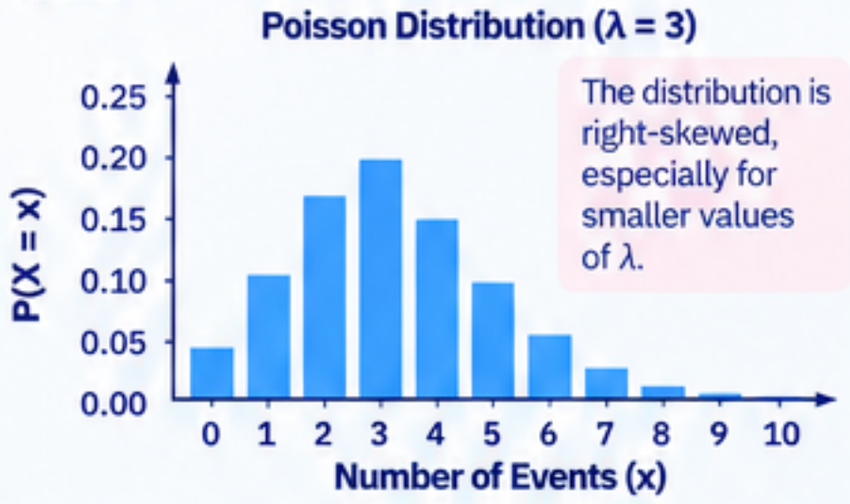
Standard Deviation:

$$\sigma = \sqrt{\lambda}$$

Properties:

- Mean equals variance.
- Distribution is right-skewed for small λ.
- As λ increases, it becomes more symmetric.

Visual Representation



For $\lambda = 3$, the highest probability is around 2 or 3 events, and the probability decreases as the number of events increases.

Example 1 – Customer Calls

A call center receives an average of 4 calls per hour ($\lambda = 4$). Find the probability of receiving exactly 2 calls in an hour.

$$P(X = 2) = \frac{e^{-4}4^{2}}{2!} = \frac{0.0183 \times 16}{2} = 0.1465$$

There is a **14.65%** chance of receiving exactly 2 calls in an hour.

Example 2 – Defective Products

A manufacturing machine produces on average 1 defective item per 100 units ($\lambda = 1$). Find the probability of getting no defective items in a batch of 100 units.

$$P(X = 0) = \frac{e^{-1}1^{0}}{0!} = e^{-1} = 0.3679$$

There is a **36.79%** chance of getting no defective items in a batch.

Example 3 – Arrivals

A bank receives an average of 2 customers per 10 minutes ($\lambda = 2$). Find the probability of receiving at least 3 customers in 10 minutes.

$$\begin{aligned} P(X \geq 3) &= 1 - P(X \leq 2) \\ &= 1 - [P(0) + P(1) + P(2)] \\ &= 1 - \left[e^{-2}\frac{2^{0}}{0!} + e^{-2}\frac{2^{1}}{1!} + e^{-2}\frac{2^{2}}{2!}\right] \\ &= 1 - [0.1353 + 0.2707 + 0.2707] \\ &= 1 - 0.6767 = 0.3233 \end{aligned}$$

There is a **32.33%** chance of receiving at least 3 customers in 10 minutes.

Excel Implementation

We can calculate Poisson probabilities in Excel using the POISSON.DIST function.

x (Events)	Formula in Excel	P(X = x) (λ = 3)
0	=POISSON.DIST(0,3,FALSE)	0.0498
1	=POISSON.DIST(1,3,FALSE)	0.1494
2	=POISSON.DIST(2,3,FALSE)	0.2240
3	=POISSON.DIST(3,3,FALSE)	0.2240
4	=POISSON.DIST(4,3,FALSE)	0.1680
5	=POISSON.DIST(5,3,FALSE)	0.1008

Excel Syntax:
=POISSON.DIST(x, λ, FALSE)

Common Calculations:

P(X ≤ k):
=POISSON.DIST(k, λ, TRUE)

P(X ≥ k):
=1-POISSON.DIST(k-1, λ, TRUE)

Common Interview Concepts

1. What is a Poisson distribution?
2. When is the Poisson distribution used?
3. State the assumptions of the Poisson distribution.
4. What are the mean and variance of a Poisson distribution?
5. How is Poisson distribution different from Binomial distribution?
6. Give a real-life example of a Poisson distribution.
7. How do you calculate P(X ≥ k) in a Poisson distribution?
8. When can we use a normal approximation for a Poisson distribution?

MODULE 5

10. Normal Distribution in Depth

Properties, Z-Scores, Standard Normal, Probability Calculations, and Applications

What is Normal Distribution?

The normal distribution is a continuous probability distribution that is symmetric and bell-shaped, defined by a mean (μ) and standard deviation (σ). It shows how values are distributed around the mean.

> *Many natural and real-world phenomena, such as height, weight, test scores, measurement errors, and financial returns, follow a normal distribution.*

Key Points

- Also called Gaussian distribution.
- Completely defined by two parameters: mean (μ) and standard deviation (σ).
- Symmetric, bell-shaped curve.
- Mean = Median = Mode.
- Total area under the curve = 1.
- Approaches the x-axis but never touches it.
- Widely used in statistics, quality control, finance, healthcare, and machine learning.

Properties of Normal Distribution

1. Symmetric around the mean (μ).
2. Mean = Median = Mode.
3. Total area under the curve = 1.
4. The curve is bell-shaped.
5. Defined by two parameters: μ (mean) and σ (standard deviation).
6. Inflection points at $\mu - \sigma$ and $\mu + \sigma$.
7. Asymptotic to the x-axis (never touches).
8. The curve is completely determined by μ and σ.

Probability Density Function (PDF)

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Where:

- x = value of the random variable
- μ = mean
- σ = standard deviation ($\sigma > 0$)
- e = 2.718 (Euler's number)
- π = 3.1416

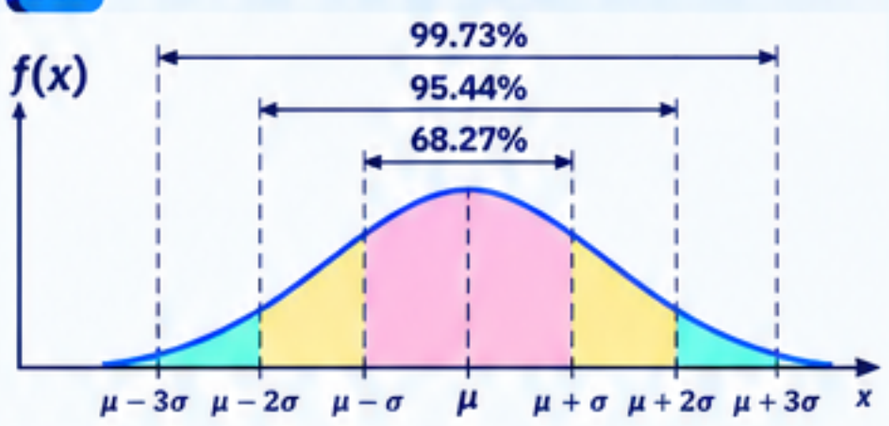
Standard Normal Distribution

If a random variable $X \sim N(\mu, \sigma^2)$, we can standardize it using Z-score:

$$Z = \frac{X - \mu}{\sigma}$$

- The standardized variable Z follows a standard normal distribution: $Z \sim N(0, 1)$
- Mean = 0, Standard deviation = 1
- We use Z-tables or Excel functions to find probabilities.

Bell Curve with Standard Deviation



Empirical Rule (68–95–99.7 Rule):

- 68.27% of data lies within 1 standard deviation ($\mu \pm 1\sigma$)
- 95.44% of data lies within 2 standard deviations ($\mu \pm 2\sigma$)
- 99.73% of data lies within 3 standard deviations ($\mu \pm 3\sigma$)

Finding Probabilities (Using Z)

To find probabilities, convert X to Z and use the standard normal distribution.

$$Z = \frac{X - \mu}{\sigma}$$

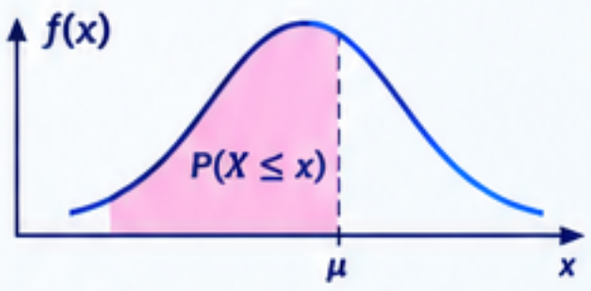
Example:

Find $P(X \leq 70)$ for $X \sim N(\mu = 65, \sigma = 5)$

$$Z = \frac{70 - 65}{5} = 1$$

$$P(X \leq 70) = P(Z \leq 1) = 0.8413$$

Area Under the Normal Curve

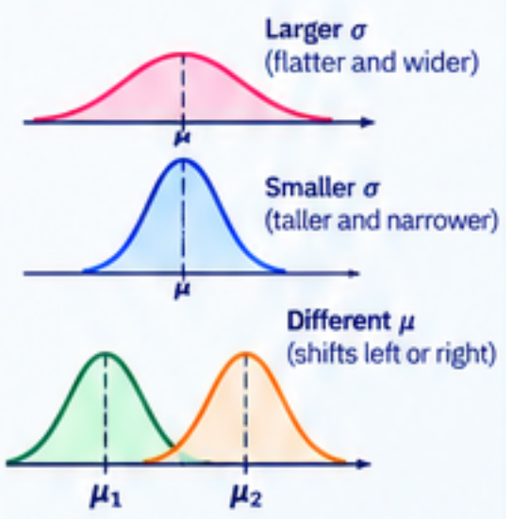


- Area represents probability.
- Total area under the curve = 1.
- Use Z-tables or Excel (NORM.DIST) for calculations.

Common Probability Calculations

1. $P(X \leq x)$ = Cumulative probability (to the left of x)
2. $P(X \geq x) = 1 - P(X \leq x)$
3. $P(a \leq X \leq b) = P(X \leq b) - P(X \leq a)$
4. $P(X \geq x) = P(Z \geq z)$
5. $P(X \leq x) = P(Z \leq z)$
6. $P(a \leq X \leq b) = P(z_1 \leq Z \leq z2)$

Shape Variations



Example 1 – Height of Individuals

Heights of adults are normally distributed with μ = 170 cm and σ = 10 cm.
Find the probability that a person's height is between 160 cm and 180 cm.

Step 1: Convert to Z-scores

$$z_1 = \frac{160 - 170}{10} = -1$$

$$z_2 = \frac{180 - 170}{10} = 1$$

Step 2: Use Z-table or Excel

$$P(-1 \leq Z \leq 1) = 0.8413 - 0.1587 = 0.6826$$

There is a **68.26%** chance that a person's height is between 160 cm and 180 cm.

Example 2 – Product Quality

The weight of a product is normally distributed with μ = 500 g and σ = 20 g.
Find the probability that a product weighs more than 540 g.

Step 1: Convert to Z-score

$$z = \frac{540 - 500}{20} = 2$$

Step 2: Use Z-table or Excel

$$P(X > 540) = P(Z > 2) = 1 - P(Z \leq 2) = 1 - 0.9772 = 0.0228$$

There is a **2.28%** chance that a product weighs more than 540 g.

Excel Implementation

In Excel, we can use NORM.DIST function.

Requirement	Excel Formula
PDF value	=NORM.DIST(x, μ, σ, FALSE)
$P(X \leq x)$	=NORM.DIST(x, μ, σ, TRUE)
$P(X \geq x)$	=1 – NORM.DIST(x, μ, σ, TRUE)
$P(a \leq X \leq b)$	=NORM.DIST(b, μ, σ, TRUE) – NORM.DIST(a, μ, σ, TRUE)
Standardize (Find Z)	=(x − μ) / σ

Standard Normal (μ = 0, σ = 1):

=NORM.S.DIST(z, TRUE) ($P(Z \leq z)$)
=NORM.S.DIST(z, FALSE) (PDF value)

Real-World Applications

- Heights and weights of people in a population.
- Test scores and exam performance.
- Measurement errors in manufacturing.
- Blood pressure, cholesterol levels in healthcare.
- Stock returns in finance.
- Delivery times, waiting times in operations.

Common Interview Concepts

1. Why is the normal distribution important?
2. What does the Empirical Rule mean?
3. How do you calculate probabilities?
4. What happens if σ increases?
5. When is normal approximation used?
6. Give real-life examples of normal distribution.

MODULE 5

11. Standard Normal Distribution

Z-Distribution, Z-Scores, Probability Calculations and Applications

What is Standard Normal Distribution?

The standard normal distribution is a normal distribution with mean $(\mu) = 0$ and standard deviation $(\sigma) = 1$.
It is denoted by $Z \sim N(0, 1)$ and is used to standardize any normal distribution so that probabilities can be easily calculated using Z-table.

> *Any normal distribution can be converted to a standard normal distribution using Z-score, which tells us how many standard deviations a value is from the mean.*

Key Points

- $Z \sim N(0, 1)$ (mean = 0, standard deviation = 1).
- Symmetric and bell-shaped curve.
- Total area under the curve = 1.
- Mean = Median = Mode = 0.
- Used to standardize any normal distribution: $Z = (X - \mu)/\sigma$.
- Z-scores tell us how far a value is from the mean in terms of standard deviations.
- Probabilities are found using Z-table or Excel (NORM.S.DIST).
- Widely used in hypothesis testing, quality control, finance, healthcare, and many real-world applications.

Properties of Standard Normal Distribution

1. Mean $(\mu) = 0$
2. Standard deviation $(\sigma) = 1$
3. Symmetric about zero
4. Bell-shaped curve
5. Total area under the curve $= 1$
6. 50% area lies to the left and 50% to the right of 0.
7. Follows the empirical rule (68–95–99.7).
8. Used as a reference distribution to find probabilities for any normal distribution.

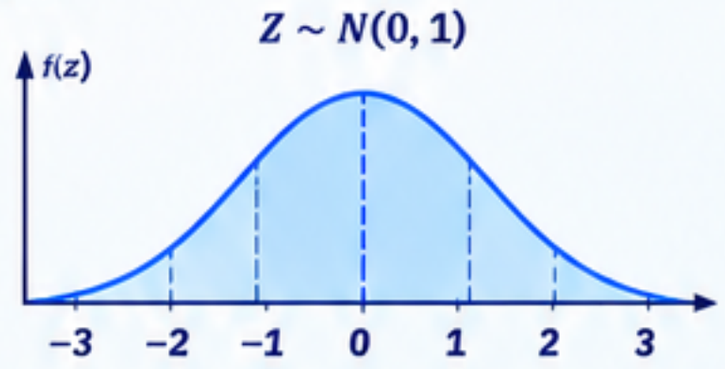
Standardization (Z-Score Formula)

$$Z = \frac{X - \mu}{\sigma}$$

Where:

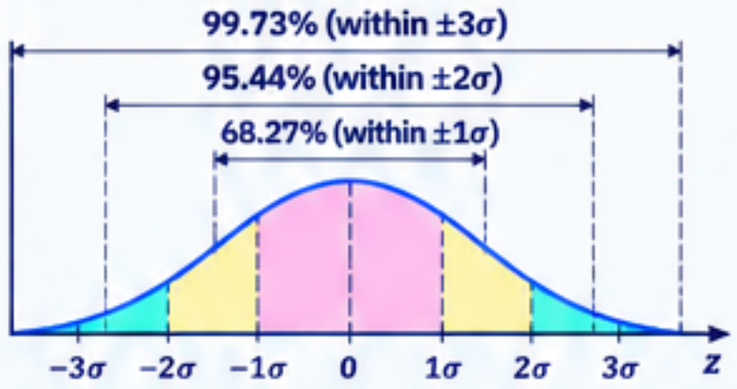
- X = value of the random variable
- μ = mean of the original normal distribution
- σ = standard deviation of the original normal distribution
- Z = standard normal variable $(Z \sim N(0, 1))$

Standard Normal Curve



Mean = 0, Standard Deviation = 1
Total area under the curve = 1
Symmetric about 0

Empirical Rule for Standard Normal Distribution



- 68.27% of values lie between −1 and +1
- 95.44% of values lie between −2 and +2
- 99.73% of values lie between −3 and +3

Finding Probabilities using Z-Table

Z-table gives the cumulative probability $P(Z \le z)$.

z	P(Z ≤ z)
−2.0	0.0228
−1.0	0.1587
0.0	0.5000
1.0	0.8413
2.0	0.9772

For negative z, use symmetry:
$P(Z \le -z) = 1 - P(Z \le z)$

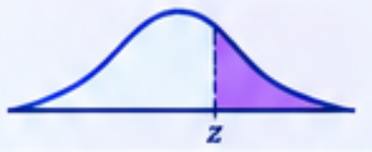
Types of Probability Calculations

1 Left-tail Probability

$P(Z \le z)$

2 Right-tail Probability

$P(Z \ge z) = 1 - P(Z \le z)$

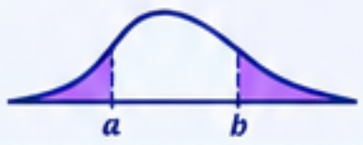


3 Between Two Values

$P(a \le Z \le b) = P(Z \le b) - P(Z \le a)$

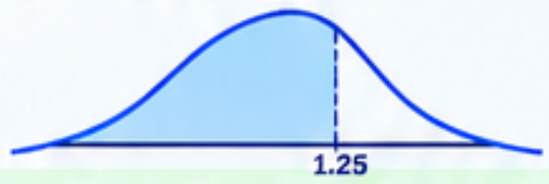
4 Outside Two Values

$P(Z \le a \text{ or } Z \ge b) = 1 - P(a \le Z \le b)$



Example 1 – Single Value Probability

Find $P(Z \le 1.25)$ using Z-table.



From Z-table: $P(Z \le 1.25) = \mathbf{0.8944}$

Example 2 – Between Two Values

Find $P(-1.0 \le Z \le 1.0)$.

$$P(-1.0 \le Z \le 1.0) = P(Z \le 1.0) - P(Z \le -1.0)$$
$$= 0.8413 - 0.1587 = 0.6826$$

Example 3 – Real World Problem

The heights of adult males are normally distributed with $\mu = 170$ cm and $\sigma = 10$ cm. Find the probability that a person's height is greater than 180 cm.

Step 1: Convert to Z-score

$$Z = \frac{180 - 170}{10} = 1$$

Step 2: Find probability

$$P(X > 180) = P(Z > 1)$$
$$= 1 - P(Z \le 1)$$
$$= 1 - 0.8413 = 0.1587$$

There is a **15.87%** chance that a person's height is greater than 180 cm.

Excel Implementation

We can use NORM.S.DIST function in Excel.

Description	Formula	Example
$P(Z \le z)$	=NORM.S.DIST(z, TRUE)	=NORM.S.DIST(1.25, TRUE) Output: 0.8944
$P(Z \ge z)$	=1 – NORM.S.DIST(z, TRUE)	=1 – NORM.S.DIST(1.25, TRUE) Output: 0.1056
$P(a \le Z \le b)$	=NORM.S.DIST(b, TRUE) – NORM.S.DIST(a, TRUE)	=NORM.S.DIST(1, TRUE) – NORM.S.DIST(–1, TRUE) Output: 0.6826

Important Notes

- Z-table gives $P(Z \le z)$ (left-tail probability).
- For right-tail, use $1 - P(Z \le z)$.
- For negative z, use symmetry: $P(Z \le -z) = 1 - P(Z \le z)$.
- Excel function NORM.S.DIST(z, TRUE) gives cumulative probability.

Common Interview Concepts

1. What is standard normal distribution?
2. Why do we use Z-score?
3. How is standard normal distribution related to normal distribution?
4. How do you find probabilities using Z-table?
5. What is the empirical rule?
6. How do you find $P(Z > z)$, $P(Z < z)$ and $P(a \le Z \le b)$?
7. What happens to the shape of the curve if we standardize a normal distribution?
8. Give real-life examples where standard normal distribution is used.

MODULE 5

12. Properties of Normal Distribution

Key Characteristics, Shape, Empirical Rule, Standardization and Applications

What are the Properties of Normal Distribution?

The normal distribution has several important properties that define its shape, behavior, and usefulness. These properties help us understand why it is widely used in statistics and real-world applications.

> *Knowing the properties of normal distribution helps us to work with data more effectively, make better decisions, and apply statistical methods correctly.*

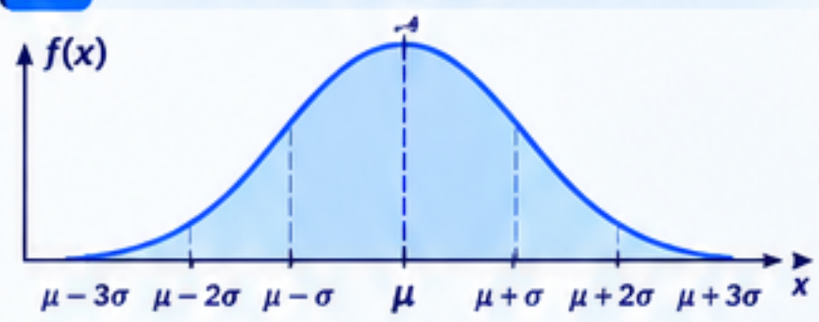
Key Points

1. Symmetric and bell-shaped curve.
2. Mean = Median = Mode.
3. Unimodal (single peak).
4. Completely defined by two parameters: μ (mean) and σ (standard deviation).
5. Total area under the curve = 1.
6. Asymptotic to the x-axis (never touches).
7. Inflection points at $\mu - \sigma$ and $\mu + \sigma$.
8. Follows the Empirical Rule (68–95–99.7).
9. Can take any real value ($-\infty$ to $+\infty$).
10. Widely used due to the Central Limit Theorem.

List of Properties

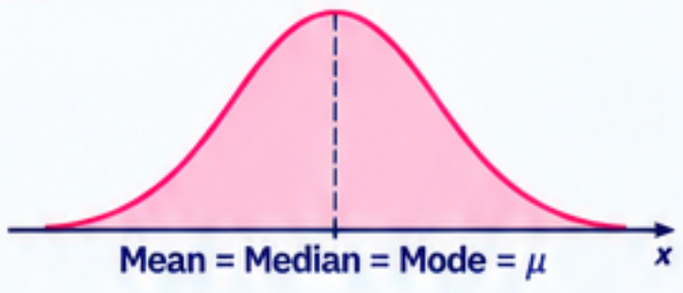
1. It is a continuous distribution.
2. It is symmetric about the mean (μ).
3. Mean = Median = Mode.
4. It is unimodal (single peak).
5. It is completely defined by two parameters: μ and σ.
6. The total area under the curve is equal to 1.
7. The curve is asymptotic to the x-axis.
8. Inflection points occur at $\mu - \sigma$ and $\mu + \sigma$.
9. It follows the Empirical Rule (68–95–99.7).
10. It can take any real value from $-\infty$ to $+\infty$.

1. Symmetric and Bell-Shaped



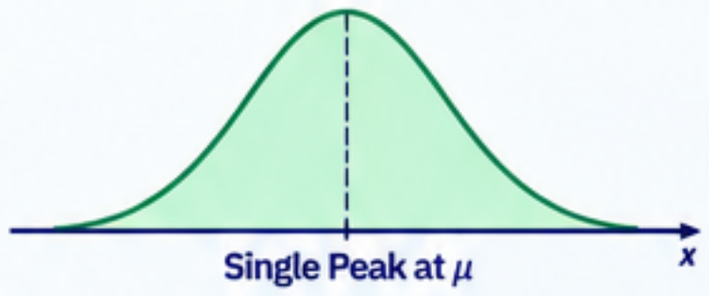
- The normal distribution is symmetric around the mean (μ).
- Both sides of the mean are mirror images.
- The curve is bell-shaped and smooth.

2. Mean = Median = Mode



- The three measures of central tendency are equal.
- This happens due to the symmetry of the distribution.
- The highest point (peak) occurs at μ.

3. Unimodal Distribution



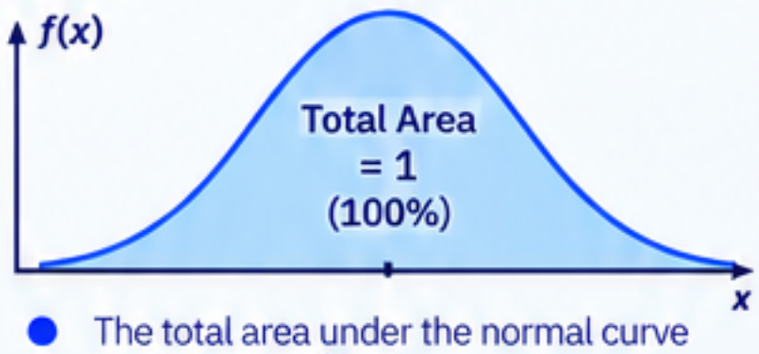
- A normal distribution has only one peak (unimodal).
- The maximum probability (highest point) occurs at the mean.

4. Defined by Two Parameters

$$X \sim N(\mu, \sigma^2)$$

- A normal distribution is completely determined by:
  - μ = mean (controls the center)
  - σ = standard deviation (controls the spread)
- Once μ and σ are known, the entire curve is fixed.

5. Total Area Under the Curve = 1

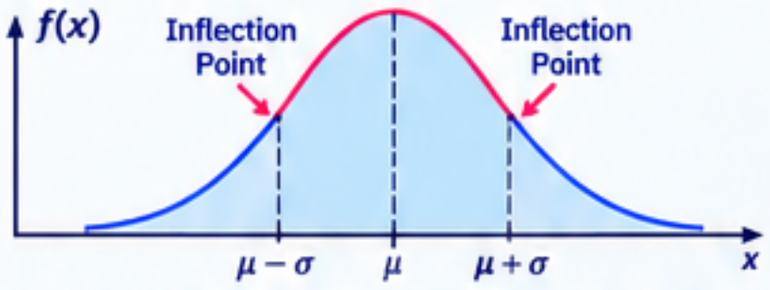


- The total area under the normal curve is always equal to 1.
- This represents the total probability (100%).

6. Asymptotic to the x-axis

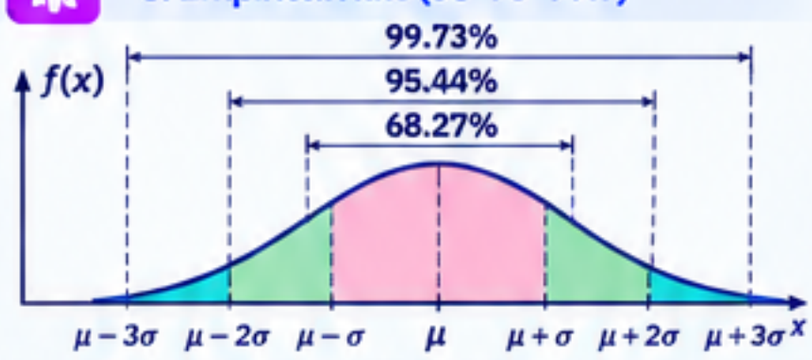
- The normal curve approaches the x-axis as it goes to $-\infty$ or $+\infty$, but it never touches it.
- This means the probability is very small in the tails but never exactly zero.

7. Inflection Points



- The inflection points occur at $\mu - \sigma$ and $\mu + \sigma$.
- At these points, the curve changes its curvature from concave up to concave down.
- These points mark the transition in the shape of the curve.

8. Empirical Rule (68–95–99.7)



- 68.27% of values lie within $\mu \pm 1\sigma$
- 95.44% of values lie within $\mu \pm 2\sigma$
- 99.73% of values lie within $\mu \pm 3\sigma$

9. Continuous and Unbounded

- The normal distribution is a continuous distribution.
- It can take any real value from $-\infty$ to $+\infty$.
- This makes it suitable for modeling variables like height, weight, test scores, etc.

10. Connection to Central Limit Theorem

- Many real-world data, when sampled repeatedly, follow an approximate normal distribution due to the Central Limit Theorem.
- This is one of the reasons why the normal distribution is widely used in statistics, business, and data analytics.

Real-World Examples

1. Heights and weights of people
2. Test scores in exams
3. Measurement errors in manufacturing
4. Product quality (e.g., diameter, thickness)
5. Financial returns (often approximately normal)

Common Interview Concepts

1. State the properties of a normal distribution.
2. Why is the normal distribution symmetric?
3. Why do mean, median and mode coincide?
4. What is the Empirical Rule?
5. Why is the normal curve asymptotic to the x-axis?
6. Give real-life examples where normal distribution is used.

MODULE 5

13. Sampling Distribution (Introduction)

Concept, Importance, Sampling Distribution of the Sample Mean (σ Known & Unknown)

What is Sampling Distribution?

A sampling distribution is the probability distribution of a sample statistic (such as the sample mean) obtained from all possible samples of a fixed size drawn from a population.

> *Instead of looking at just one sample, we look at the distribution of a statistic across many possible samples.*

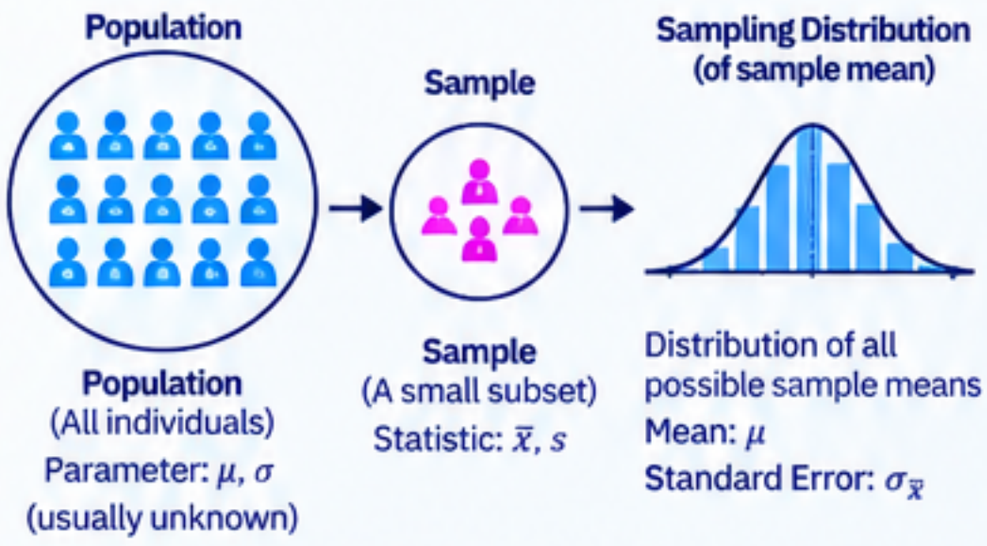
Key Points

- Formed by taking all possible samples of the same size from a population.
- Each sample gives a sample statistic (e.g., sample mean).
- Shows how the sample statistic varies from sample to sample.
- Helps us understand sampling variability (randomness in samples).
- The shape, mean, and standard error depend on the population distribution and sample size.
- Widely used in hypothesis testing, confidence intervals, and estimation.

Why is it Important?

1. Allows us to make inferences about a population using sample data.
2. Helps in estimating population parameters (μ, σ).
3. Forms the basis for confidence intervals and hypotbesis testing.
4. Explains sampling variability.
5. According to the Central Limit Theorem, the sampling distribution of the sample mean is approximately normal for large sample sizes.

Population vs Sample vs Sampling Distribution



Sampling Distribution of the Sample Mean

If the population has mean μ and standard deviation σ, then the sampling distribution of the sample mean $\bar{x}$ has:

$$\mu_{\bar{x}} = \mu \qquad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

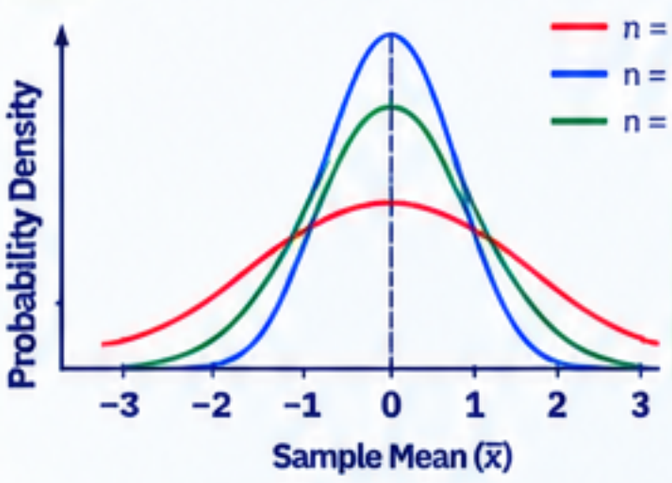
Where:

- $\mu_{\bar{x}}$ = mean of the sampling distribution of $\bar{x}$ (equal to population mean)
- $\sigma_{\bar{x}}$ = standard error of the mean
- σ = population standard deviation
- n = sample size

Key Properties

1. Mean of $\bar{x}$ is equal to the population mean: $\mu_{\bar{x}} = \mu$.
2. Standard deviation of $\bar{x}$ is the standard error: $\sigma_{\bar{x}} = \sigma/\sqrt{n}$.
3. The distribution becomes narrower as sample size (n) increases.
4. If the population is normally distributed, the sampling distribution of $\bar{x}$ is also normal (for any n).
5. If the population is not normal, by the Central Limit Theorem, the sampling distribution of $\bar{x}$ is approximately normal for large n ($n \geq 30$).

Visual Representation



As sample size increases, the sampling distribution becomes narrower (less variability).

Example 1 – σ Known

A population has $\mu = 50$ and $\sigma = 10$. Find the mean and standard error of the sampling distribution of $\bar{x}$ for $n = 25$.

Solution:

- Mean: $\mu_{\bar{x}} = \mu = 50$
- Standard Error: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{25}} = \frac{10}{5} = 2$

$$So, \bar{x} \sim N(50, 2^2)$$

Example 2 – σ Unknown (Using s)

A sample of size $n = 36$ is taken from a population (σ unknown). The sample standard deviation is $s = 12$. Find the standard error of the sample mean.

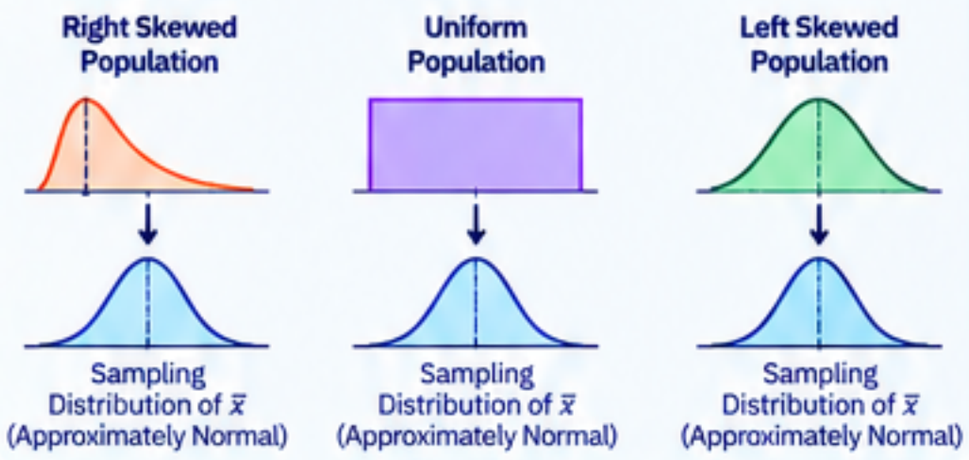
Solution:

- Standard Error: $\sigma_{\bar{x}} \approx \frac{s}{\sqrt{n}} = \frac{12}{\sqrt{36}} = \frac{12}{6} = 2$

Even when σ is unknown, we use the sample standard deviation (s) to estimate the standard error of the mean.

Central Limit Theorem (Connection)

If the population distribution is not normal, the sampling distribution of $\bar{x}$ will still be approximately normal for large sample sizes (usually $n \geq 30$), regardless of the shape of the population distribution.



Excel Implementation (Simulation)

We can simulate the sampling distribution of the mean in Excel:

1. Generate a population (e.g., normal or non-normal).
2. Take repeated random samples of size n.
3. Calculate the sample mean for each sample.
4. Plot the distribution of sample means using a histogram.
5. Observe that as n increases, the distribution becomes approximately normal and the standard error decreases.

Common Interview Concepts

1. What is a sampling distribution?
2. How is it different from a population distribution?
3. What are the mean and standard deviation of the sampling distribution of $\bar{x}$?
4. Why does the sampling distribution of $\bar{x}$ become approximately normal for large n?
5. How does sample size affect the shape and spread of the sampling distribution?
6. What is the role of the Central Limit Theorem in sampling distributions?
7. How do we estimate the standard error when σ is unknown?
8. Give a real-life example where sampling distribution is used.

MODULE 5

14. Real World Examples & Applications

How Normal Distribution is Used in Business, Healthcare, Finance and More

What You Will Learn?

We will explore how the normal distribution is used in real-world scenarios across different industries. You will see how the concepts of mean, standard deviation, probabilities, and Z-scores help in decision-making, quality control, risk analysis, and more.

> *The normal distribution is not just a theoretical concept — it is one of the most widely used distributions in real-life business and analytical problems.*

Key Points

- Many real-world variables approximately follow a normal distribution.
- Used in business, healthcare, finance, manufacturing, education, and more.
- Helps in prediction, decision-making, risk analysis, and quality control.
- Allows calculation of probabilities using Z-scores.
- Based on the Central Limit Theorem, many sample means are normally distributed.
- Used for benchmarking, performance evaluation, and setting standards.

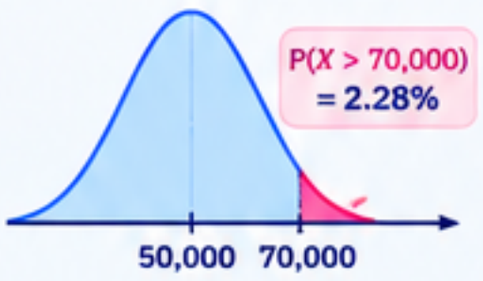
Why is it Important?

1. It helps organizations make data-driven decisions.
2. Used to set realistic targets and performance benchmarks.
3. Helps identify unusual or rare events (outliers).
4. Widely used in quality control and process improvement.
5. Essential for risk analysis, forecasting, and strategic planning.

1. Business Example – Sales Analysis

A retail company's monthly sales are normally distributed with a mean of ₹50,000 and a standard deviation of ₹10,000. What is the probability that monthly sales will be more than ₹70,000?

Sales Distribution
$\mu = 50{,}000$
$\sigma = 10{,}000$



Solution:

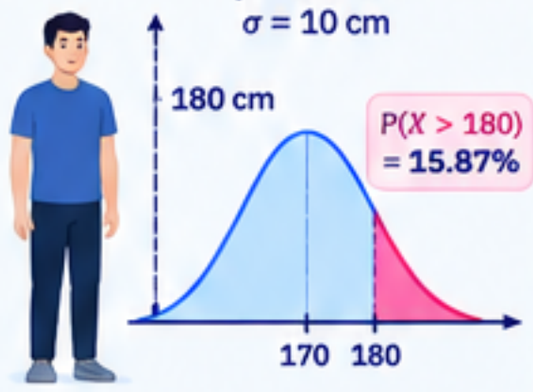
- Mean (μ) = 50,000
- Standard Deviation (σ) = 10,000
- We want $P(X > 70{,}000)$
- $Z = (70{,}000 - 50{,}000) / 10{,}000 = 2$
- From Z-table: $P(X > 70{,}000) = P(Z > 2) = 1 - 0.9772 = 0.0228$

There is a **2.28%** chance that monthly sales will exceed **₹70,000.**

2. Healthcare Example – Human Heights

Heights of adult males are normally distributed with a mean of 170 cm and a standard deviation of 10 cm. What percentage of men are taller than 180 cm?

Height Distribution (Adult Males)
$\mu = 170$ cm
$\sigma = 10$ cm



Solution:

- Mean (μ) = 170
- Standard Deviation (σ) = 10
- $Z = (180 - 170) / 10 = 1$
- $P(X > 180) = P(Z > 1) = 1 - 0.8413 = \mathbf{0.1587}$

Around **15.87%** of adult males are taller than 180 cm.

3. Manufacturing Example – Quality Control

The weight of a product is normally distributed with a mean of 500 g and a standard deviation of 20 g. What is the probability that a randomly selected product weighs less than 460 g?

Target Weight
500 g (±20 g)

Solution:

- Mean (μ) = 500
- Standard Deviation (σ) = 20
- $Z = \mathbf{(460 - 500) / 20 = -2}$
- $P(X < 460) = P(Z < -2) = 0.0228$

There is a **2.28%** chance that a product weighs less than 460 g (may be underweight/defective).

4. Finance Example – Stock Returns

Daily returns of a stock are approximately normally distributed with a mean of 0.5% and a standard deviation of 2%. What is the probability that the daily return will be less than –3%?

Stock Return Distribution
$\mu = 0.5\%$
$\sigma = 2\%$

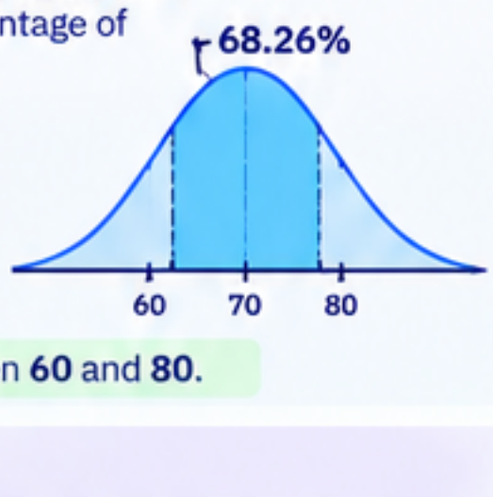
Solution:

- Mean (μ) = 0.5%
- Standard Deviation (σ) = 2%
- $Z = (-3\% - 0.5\%) / 2\% = -1.75$
- $P(X < -3\%) = P(Z < -1.75) = 0.0401$

There is a **4.01%** chance that the daily return will be less than –3%.

5. Education Example – Exam Scores

Exam scores are normally distributed with a mean of 70 and a standard deviation of 10. What percentage of students score between 60 and 80?



Solution:

- $Z_1 = (60 - 70) / 10 = -1$
- $Z_2 = (80 - 70) / 10 = 1$
- $P(60 \le X \le 80) = P(-1 \le Z \le 1) = 0.8413 - 0.1587 = 0.6826$

Around **68.26%** of students score between **60** and **80.**

6. Real Life Example – Customer Delivery Time

Delivery time of an e-commerce company is normally distributed with a mean of 3 days and a standard deviation of 1 day. What is the probability that a delivery takes more than 5 days?

Delivery Time Distribution
$\mu = 3$ days
$\sigma = 1$ day

Solution:

- Mean (μ) = 3
- Standard Deviation (σ) = 1
- $Z = (5 - 3) / 1 = 2$
- $P(X > 5) = P(Z > 2) = 0.0228$

There is a **2.28%** chance that a delivery takes more than 5 days.

Common Interview Questions

1. Give real-world examples where normal distribution is used.
2. How is normal distribution used in quality control?
3. Why is normal distribution important in business analytics?
4. Can you give an example from healthcare or finance?
5. How does the normal distribution help in risk management?

Key Takeaways

- The normal distribution is widely used in real-world problems.
- Helps in decision-making, forecasting, risk analysis, and quality control.
- Applicable in multiple industries: business, healthcare, finance, education, manufacturing, etc.
- Probabilities and Z-scores allow us to quantify and understand uncertainty.
- A small probability in the tail often indicates a rare or unusual event (outlier).

MODULE 5

15. More on Real World Applications

Advanced Use Cases of Normal Distribution in Business, Healthcare, Finance & More

What You Will Learn?

In this lecture, we will explore more real-world applications of the normal distribution across different industries. You will see how normal distribution helps in quality control, risk management, demand forecasting, performance analysis, and decision-making.

> *The normal distribution is one of the most practical tools in the real world because many natural and business processes follow a normal pattern.*

Key Points

- Normal distribution is widely used in real-world decision-making.
- Helps in measuring probability of events.
- Used for setting benchmarks and targets.
- Helps in quality control, risk analysis, forecasting, and performance measurement.
- Based on mean (μ) and standard deviation (σ).
- Z-scores help in comparing values and standardizing data.
- Applicable across multiple industries like manufacturing, healthcare, finance, education, and e-commerce.

Why It Matters?

1. Helps businesses make data-driven decisions.
2. Used to identify unusual or rare events (outliers).
3. Helps in setting realistic targets and performance benchmarks.
4. Widely used in risk management and forecasting.
5. Essential for improving quality, reducing costs, and optimizing operations.

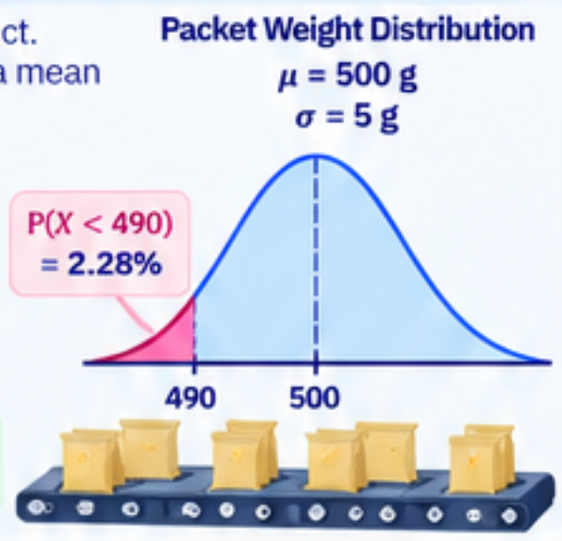
1. Manufacturing – Quality Control

A machine fills packets with 500 g of product. The weights are normally distributed with a mean of 500 g and a standard deviation of 5 g. What percentage of packets will weigh less than 490 g?

Solution:

- $\mu = 500, \ \sigma = 5$
- $Z = (490 - 500) / 5 = -2$
- $P(X < 490) = P(Z < -2) = 0.0228$

About **2.28%** of packets will weigh less than 490 g (underfilled).



2. Healthcare – Blood Pressure Analysis

Systolic blood pressure of adults is normally distributed with a mean of 120 mmHg and a standard deviation of 15 mmHg. What percentage of adults have blood pressure greater than 150 mmHg?

Solution:

- $\mu = 120, \ \sigma = 15$
- $Z = (150 - 120) / 15 = 2$
- $P(X > 150) = P(Z > 2) = 0.0228$

About **2.28%** of adults have blood pressure greater than 150 mmHg.

3. Finance – Investment Returns

Monthly returns of a stock are normally distributed with a mean of 1% and a standard deviation of 5%. What is the probability of a negative return (less than 0%)?

Solution:

- $\mu = 1\%, \ \sigma = 5\%$
- $Z = (0 - 1) / 5 = -0.2$
- $P(X < 0) = P(Z < -0.2) = 0.4207$

There is a **42.07%** chance of a negative monthly return.

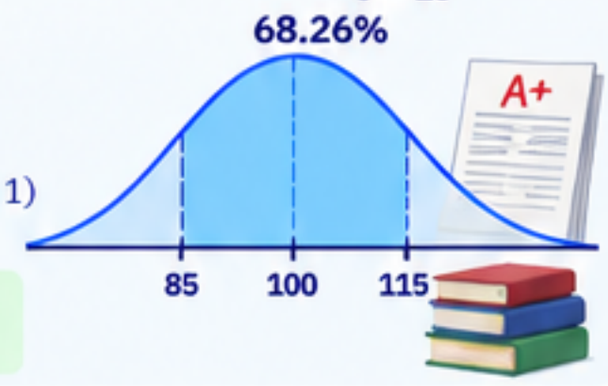
4. Education – Exam Performance

Standardized test scores are normally distributed with a mean of 100 and a standard deviation of 15. What percentage of students score between 85 and 115?

Solution:

- $Z_1 = (85 - 100) / 15 = -1$
- $Z_2 = (115 - 100) / 15 = 1$
- $P(85 \le X \le 115) = P(-1 \le Z \le 1)$
  $= 0.8413 - 0.1587 = 0.6826$

About **68.26%** of students score between 85 and 115.



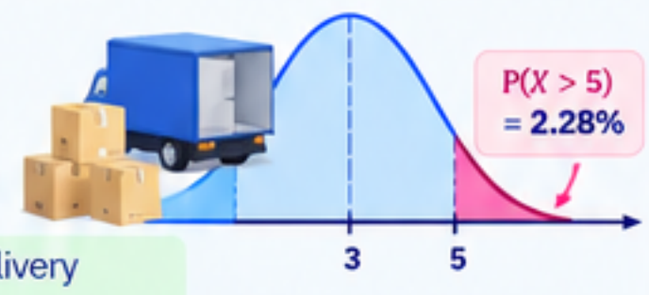
5. E-commerce – Delivery Time

Delivery time is normally distributed with a mean of 3 days and a standard deviation of 1 day. What is the probability that a delivery takes more than 5 days?

Solution:

- $\mu = 3, \ \sigma = 1$
- $Z = (5 - 3) / 1 = 2$
- $P(X > 5) = P(Z > 2) = 0.0228$

There is a **2.28%** chance that a delivery takes more than 5 days.



6. Real-World Business Decision Examples

Area	Use Case	Normal Distribution Helps In
Sales & Marketing	Predict customer spending	Estimate probability of higher or lower sales.
Manufacturing	Product quality control	Identify defective items, reduce wastage.
Healthcare	Patient health metrics	Identify risk of high/low values.
Finance	Stock returns, credit risk	Measure risk and set confidence intervals.
Education	Exam scores and grading	Set performance benchmarks.
E-commerce	Delivery time estimation	Forecast delays and improve logistics.
HR & People Analytics	Employee performance	Understand distribution of scores and set appraisal standards.

Key Takeaways

- Normal distribution has wide real-world applications.
- Helps in quality control, risk analysis, forecasting, and performance measurement.
- Z-scores make it easy to calculate probabilities.
- Used in manufacturing, healthcare, finance, education, e-commerce, and more.
- Understanding normal distribution helps in making better, data-driven decisions.

Common Interview Questions

1. Give real-world examples where normal distribution is used.
2. How is normal distribution used in quality control?
3. How is it used in risk analysis and finance?
4. Can you give an example from healthcare or education?
5. How does normal distribution help in business decision-making?
6. What happens if the data is not normally distributed?

MODULE 5

16. Interview Concepts - Normal Distribution

Concepts Summary, Important Interview Questions & Answers (For Data Analysts & Data Scientists)

1. Quick Recap – Key Concepts

1. Normal distribution is a continuous, symmetric, bell-shaped distribution.
2. Defined by two parameters: mean (μ) and standard deviation (σ).
3. Mean = Median = Mode.
4. Total area under the curve = 1 (100%).
5. 68–95–99.7 empirical rule.
6. Standard normal distribution: $Z \sim N(0, 1)$.
7. Z-score: tells how many standard deviations a value is from the mean.
8. Widely used in real-world applications and statistical inference.

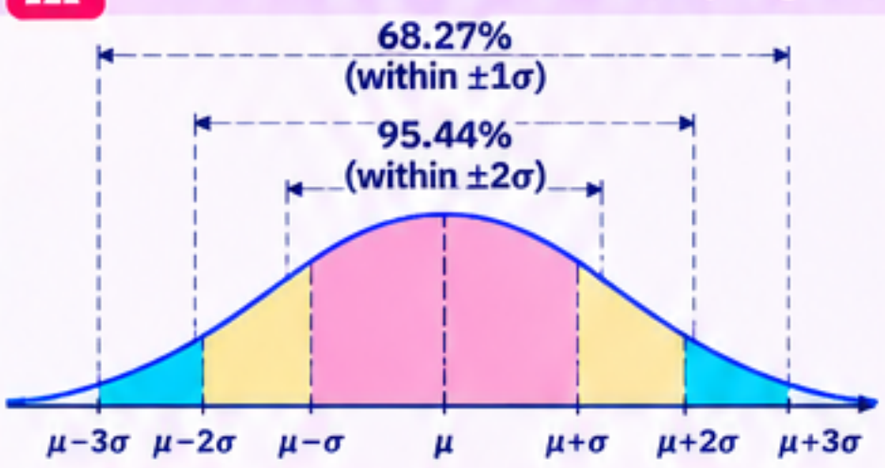
2. Important Formulas (Quick Review)

Normal Distribution	$X \sim N(\mu, \sigma^2)$
Standardization (Z-score)	$Z = \frac{X - \mu}{\sigma}$
Standard Normal	$Z \sim N(0, 1)$
Probability (Left-tail)	$P(X \le a) = P(Z \le (a - \mu)/\sigma)$
Probability (Right-tail)	$P(X \ge a) = 1 - P(Z \le (a - \mu)/\sigma)$
Between Two Values	$P(a \le X \le b) = P((a - \mu)/\sigma \le Z \le (a - \mu)/\sigma)$
Empirical Rule	68% ($\pm 1\sigma$), 95% ($\pm 2\sigma$), 99.7% ($\pm 3\sigma$)

3. Shape & Properties (Interview Points)

- Continuous probability distribution
- Symmetric about the mean
- Mean = Median = Mode
- Bell-shaped curve
- Total area under the curve = 1
- Asymptotic to the x-axis (never touches)
- Inflection points at $\mu - \sigma$ and $\mu + \sigma$
- Defined by two parameters: μ and σ
- Standard normal: mean 0, standard deviation 1
- Follows Empirical Rule (68–95–99.7)
- Widely used due to Central Limit Theorem

4. Normal Distribution Curve (Key Regions)



- 68.27% of values lie between $\mu - \sigma$ and $\mu + \sigma$
- 95.44% of values lie between $\mu - 2\sigma$ and $\mu + 2\sigma$
- 99.73% of values lie between $\mu - 3\sigma$ and $\mu + 3\sigma$

5. Standard Normal Distribution (Z)

$$Z \sim N(0, 1)$$

- Mean (μ) = 0
- Standard Deviation (σ) = 1
- Used to standardize any normal distribution
- Probabilities are found using Z-table or Excel (NORM.S.DIST)

6. Z-Scores – Interpretation

$$Z = \frac{X - \mu}{\sigma}$$

Z-score tells how many standard deviations a value is from the mean and in which direction.

Z-score	Position	Meaning
0	At mean	Value is equal to the mean
+1	1 SD above mean	Higher than 84.13% of values
−1	1 SD below mean	Higher than 15.87% of values
+2	2 SD above mean	Higher than 97.72% of values
−2	2 SD below mean	Higher than 2.28% of values

7. Top Interview Questions & Answers

1 What is a normal distribution?

Answer:
Normal distribution is a continuous, symmetric, bell-shaped distribution defined by mean (μ) and standard deviation (σ). It is widely used in statistics and real-world applications.

2 Why is normal distribution important in data analysis?

Answer:
It helps in understanding data patterns, setting benchmarks, making probabilistic predictions, performing hypothesis testing, quality control, and risk analysis. Many real-world variables approximately follow a normal distribution.

3 What do μ and σ represent?

Answer:
μ (mean) represents the central value of the distribution.
σ (standard deviation) represents the spread or variability of the data.

4 What is a Z-score?

Answer:
Z-score shows how many standard deviations a value is from the mean.

$$Z = \left(\frac{X - \mu}{\sigma}\right)\sigma$$

A positive Z means above the mean and a negative Z means below the mean.

5 What is the standard normal distribution?

Answer:
Standard normal distribution is a normal distribution with mean 0 and standard deviation 1. It is denoted as $Z \sim N(0, 1)$ and is used to standardize any normal distribution.

6 What is the Empirical Rule?

Answer:
In a normal distribution:

- 68.27% of values lie within $\pm 1\sigma$
- 95.44% of values lie within $\pm 2\sigma$
- 99.73% of values lie within $\pm 3\sigma$

7 How do you calculate probabilities using normal distribution?

Convert the value to Z-score using

$$Z = (X - \mu/\sigma$$

Then use Z-table or Excel function (NORM.S.DIST) to find the probability.

8 What is the total area under the normal curve?

The total area under the normal distribution curve is 1 (or 100%).
Probabilities represent areas under the curve.

9 Why does the normal distribution never touch the x-axis?

It is asymptotic to the x-axis, meaning the probability approaches zero but never becomes exactly zero.

10 What are inflection points?

Inflection points occur at $\mu - \sigma$ and $\mu + \sigma$, where the curve changes its concavity.

11 What is the relationship between mean, median, and mode?

In a normal distribution:
Mean = Median = Mode.

12 Can you give a real-world example of normal distribution?

Human heights, product weights, exam scores, blood pressure, and measurement errors are commonly normally distributed.

13 How is normal distribution used in hypothesis testing?

Many test statistics (e.g., Z-test) follow a normal distribution under the null hypothesis, allowing us to calculate p-values and make decisions.

14 What is the normal distribution widely used in quality control?

Because many process measurements (e.g., weight, size, strength) are approximately normal, it helps identify defective items, set control limits and improve quality.

15 What is the Central Limit Theorem and how is it related?

The Central Limit Theorem states that the sampling distribution of the sample mean is approximately normal for large sample sizes, regardless of the original distribution.

16 When would you not use normal distribution?

When the data is highly skewed, contains outliers, or is not continuous (e.g., count data, categorical data), a normal distribution may not be appropriate.

Key Takeaways (for Interview)

- Understand shape, properties, and parameters (μ, σ).
- Be comfortable with Z-scores and probability calculations.
- Know real-world applications and limitations.
- Understand connection with Central Limit Theorem.

Tips for Interview Preparation

- Practice numerical problems using Z-table and Excel.
- Be ready to explain concepts with diagrams.
- Give real-world examples from your domain.
- Understand when and why normal distribution is used.
- Highlight applications relevant to your role (analytics, ML, business).

Useful Excel Functions

Function	Purpose
NORM.DIST(x, μ, σ, TRUE)	Cumulative probability
NORM.S.DIST(z, TRUE)	Standard normal cumulative
NORM.INV(p, μ, σ)	Find value for given probability
NORM.S.INV(p)	Z-value for given probability